REVIEW

Benchmarking multi-output models for continuous music emotion prediction

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Abstract
Predicting the emotional content of music is a long-standing challenge in audio signal processing and machine learning, and although Music Emotion Recognition (MER) increasingly relies on deep learning, a systematic comparison of audio descriptor families and of their integration through ensemble strategies and fusion levels for continuous emotion prediction remains underexplored. This study addresses this gap within Russell's circumplex model of valence and arousal. Six machine learning paradigms (LSTM, MLP, Random Forest, TabNet, PLSR, XGBoost) are evaluated across three theoretically diverse descriptor families: OpenSMILE (signal-processing), PAA (psychoacoustic), and Librosa (MIR-standard). A ranking-based consistency criterion identifies the four paradigms competitive across all families, and the resulting twelve base models are combined at three fusion levels: late, early, and hybrid stacking. The framework is benchmarked against four state-of-the-art architectures: three pre-trained audio Transformers (MERT, HuBERT, wav2vec 2.0) and a Conformer trained from scratch. All experiments use 5-fold file-wise cross-validation, with Mean Euclidean Distance (MED) as the primary joint metric alongside per-dimension RMSE, MAE, and Concordance Correlation Coefficient. LSTM on OpenSMILE and on PAA achieves the strongest individual performance (MED = 0.241), and hybrid Ridge stacking attains the best overall result (MED = 0.213, an 11.4% reduction), validated by paired Wilcoxon tests. MERT remains the strongest model in absolute terms (MED = 0.201), but the proposed framework outperforms the speech-pre-trained Transformers and the from-scratch Conformer. Meta-learner weight analysis and per-feature permutation importance expose interpretable, dimension-specific contribution patterns across descriptor families, learning paradigms, and individual features, providing a reproducible and competitive non-Transformer baseline for continuous MER research.

Keywords Music emotion recognition · Valence–arousal regression · Ensemble learning · Audio descriptors · Affective computing

1 Introduction

Music has long been recognized as one of the most powerful vehicles for emotional expression and communication [1]. Unlike language, which conveys meaning through explicit symbols, music operates on a more abstract level, often evoking affective responses that are immediate, visceral, and deeply personal. A simple melodic



progression, a sudden change in rhythm, or a shift in harmony can trigger emotional reactions that resonate across individuals and cultures. This intimate connection between music and emotion has made the study of how emotions are perceived, experienced, and classified in musical contexts a longstanding focus of research at the intersection of psychology, musicology, neuroscience, and computer science.

In recent decades, the field of Music Emotion Recognition (MER) has emerged as a central area of computational music research [2]. The goal of MER is to develop methods capable of automatically identifying the emotional content or impact of music, either as perceived by listeners or as intended by composers and performers. This capability is not only of theoretical interest for understanding the cognitive and affective mechanisms behind music perception, but also of practical significance for a wide range of applications. Systems that can recognize or predict emotions in music have the potential to enhance music recommendation engines, support personalized playlists, enable adaptive gaming and virtual reality experiences, and contribute to fields as diverse as affective computing, music therapy, and human-computer interaction.

Approaches to MER typically follow two complementary paradigms. The categorical model classifies emotions into discrete labels such as happiness, sadness, anger, or calmness, aligning with everyday language but limiting the representation of gradual emotional transitions. The dimensional model, grounded in Russell's circumplex theory of affect [3], represents emotions along continuous scales of valence (from positive to negative affect) and arousal (from low to high activation), capturing subtle dynamics and complex affective blends. Both paradigms have proven valuable, and contemporary research increasingly seeks to bridge them, acknowledging that musical emotions can be both discretely labeled and continuously experienced.

Automatic emotion recognition in music is inherently challenging due to the multifaceted nature of musical signals. Audio recordings encompass features at multiple levels, ranging from low-level acoustic descriptors such as timbre, tempo and spectral shape, to mid-level constructs such as harmony, rhythm and melody, and to higher-level cultural cues. At the same time, emotional perception is strongly modulated by listener-dependent factors including personality, cultural background, and situational context. This variability complicates the design of computational models that generalize across listeners and musical styles.

Machine learning advances have transformed MER from handcrafted features paired with traditional classifiers to deep learning architectures capable of learning representations directly from audio. Recent approaches encompass convolutional, recurrent, and transformer-based models that capture both local and long-range temporal dependencies in emotional expression.

Despite these advances, fundamental challenges persist. The subjectivity of emotional perception, given that the same music elicits different responses across listeners and contexts, complicates ground truth annotation and model evaluation. Continuous valence–arousal annotations, while enabling fine-grained analysis, are often noisy and inconsistent. Furthermore, MER systems frequently struggle to generalize across datasets and musical styles, highlighting ongoing needs for robust modeling strategies and domain adaptation approaches.

The present work addresses these challenges through a comprehensive comparative framework for continuous valence–arousal prediction. Rather than proposing novel architectures, we systematically investigate how different families of audio descriptors, representing signal-processing, psychoacoustic, and music information retrieval perspectives, compare in their ability to capture emotional trajectories, how their complementary strengths can be leveraged through principled ensemble strategies, and how the resulting predictions can be interpreted at the level of both descriptor families and individual features. The contributions of this work can be summarized as follows. First, we provide a systematic comparison of three theoretically diverse audio descriptor families, namely OpenSMILE (signal-processing), PAA (psychoacoustic, computed via Zwicker's models), and Librosa (MIR-standard), evaluated under a unified preprocessing and 5-fold file-wise cross-validation protocol. Second, we assess six machine-learning paradigms (LSTM, MLP, Random Forest, TabNet, PLSR, XGBoost) in a multi-output configuration that jointly predicts valence and arousal, yielding 18 descriptor–method combinations as base learners. Third, we integrate the four most consistent paradigms (LSTM, MLP, Random Forest, XGBoost) across the three descriptor families, for a total of 12 base models, through three late-fusion ensemble strategies:

arithmetic averaging, Leave-One-Fold-Out (LOFO) weighted averaging, and Ridge meta-learning stacking. Fourth, we extend the integration beyond decision-level fusion by adding an early-fusion stage on the concatenated 1085-dimensional feature space and a hybrid Ridge stacking meta-learner that combines the early- and late-fusion predictions, allowing a direct comparison of the three canonical fusion levels on the same task and splits. Fifth, we develop a two-level interpretability analysis of the framework: at the family level, a dimension-specific weight analysis of the Ridge meta-learner exposes which descriptor families and modeling paradigms contribute most to each emotional axis; at the feature level, a per-feature permutation-importance analysis is computed separately for each descriptor family, identifying which specific descriptors within OpenSMILE, PAA, and Librosa carry the predictive content and how concentrated or dispersed this content is across each pool. To the best of our knowledge, this is the first work that combines descriptor-family weight analysis with per-feature permutation importance in a continuous MER setting, providing interpretability at two complementary levels of granularity. Sixth, we benchmark the proposed framework against four state-of-the-art audio architectures: three pre-trained Transformers (MERT, HuBERT, wav2vec 2.0) under a unified multi-layer weighted-sum probing protocol and a Conformer trained from scratch on the DEAM mel-spectrograms; the ensemble gains are validated through paired Wilcoxon signed-rank significance tests. Together, these elements establish a reproducible, interpretable, and transparent baseline for future research in continuous music emotion recognition.

At the structural level, the remainder of this article is organized as follows. Section 2 reviews the most relevant literature on Music Emotion Recognition, with particular attention to continuous prediction approaches based on the valence–arousal model. Section 3 describes the dataset and annotation scheme adopted in this study, while Sect. 4 provides an overview of the general methodological framework and data preprocessing pipeline. Section 5 focuses on the acoustic descriptors employed, outlining their characteristics and feature extraction procedures. Section 6 presents the set of machine learning methods evaluated, spanning deep, ensemble-based, and linear paradigms. Section 7 introduces the ensemble learning strategies designed to integrate the outputs of individual models, including the late-fusion ensembles built on the base learners, the early-fusion stage operating on concatenated descriptors, and the hybrid stacking meta-learner that combines early and late predictions. Section 8 details the experimental setup, including the training protocol, evaluation metrics, and implementation details. Section 9 reports and discusses the experimental results: individual base models and ensemble configurations are first analysed in terms of Mean Euclidean Distance and Concordance Correlation Coefficient, supported by paired Wilcoxon signed-rank significance tests; a dedicated subsection then compares the proposed framework against four state-of-the-art audio architectures, namely three pre-trained Transformers (MERT, HuBERT, wav2vec 2.0) and a Conformer trained from scratch on the DEAM mel-spectrograms; meta-learner weight analysis closes the section, exposing which descriptor families and modeling paradigms contribute most to each emotional axis, together with a per-feature permutation-importance analysis that identifies the most informative descriptors within each family. Finally, Sect. 10 summarizes the main findings, discusses their broader implications, and outlines directions for future research in continuous music emotion prediction.

2 State of the art

Research in Music Emotion Recognition (MER) has made remarkable progress in the last two decades, supported by the creation of dedicated datasets and the rapid evolution of machine learning methods. A central role in this development has been played by annotated corpora that provide systematic ground truth for model training and evaluation. The DEAM dataset [4], for example, is one of the most widely used resources, offering dynamic annotations of valence and arousal over 1,800 music excerpts and serving as the benchmark in multiple MediaEval evaluation campaigns. Similarly, PMEmo [5] has enriched the MER landscape by combining continuous valence–arousal annotations with electrodermal activity (EDA) recordings from listeners, thus enabling multimodal studies that integrate audio and physiological signals. MERP [6] contributes another valuable dimension by collecting dynamic ratings together with rater profile information, paving the way for research on personalization

and bias in emotion perception. Other corpora address the subjectivity and variability of emotion annotation in different ways: MUSAV [7] employs relative pairwise judgments of arousal and valence instead of absolute ratings to mitigate inconsistencies, while TROMPA-MER [8] uses a citizen science platform to gather annotations enriched with demographic, linguistic, and contextual metadata. Additional datasets such as MER1101 [9] and those proposed in multilingual settings [10] further expand the experimental basis for MER research, providing diversity both in music styles and in annotator populations.

Methodologically, early approaches relying on handcrafted features such as timbre descriptors, tempo, or harmonic features have given way to deep learning pipelines capable of automatically extracting hierarchical representations from raw signals. Convolutional neural networks (CNNs) applied to spectrograms or log-Mel representations are among the most common approaches [9, 11], often augmented by multi-scale structures and attention modules that capture emotional cues at different temporal and spectral levels. Recurrent neural networks (RNNs) and their variants, such as GRUs and LSTMs, have been widely employed to model temporal dependencies and dynamic changes in affect [12, 13], sometimes incorporating structured label spaces to reduce ambiguity in emotional categories. More recent end-to-end architectures operate directly on raw audio waveforms, combining convolutional feature extractors with autoencoder-inspired reconstruction layers and bidirectional recurrent networks, showing particular improvements in predicting valence trajectories [13]. Hybrid designs that integrate efficient convolutional backbones with temporal modules have also been proposed, aiming to model both local acoustic detail and long-range emotional flow [11, 14]. Beyond convolutional and recurrent topologies, graph-based architectures have recently emerged as a promising alternative: Zhang et al. [15] proposed a multi-scale controllable graph convolutional network that operates on spectro-temporal graphs derived from the audio signal, achieving competitive performance on continuous valence–arousal prediction by explicitly modelling relational structure across spectro-temporal regions. In parallel, ensemble learning strategies [16] have been explored to enhance robustness by aggregating predictions across diverse models, while representation learning approaches based on triplet neural networks [17] have been adapted to regression, allowing emotion-aware embeddings to be learned and subsequently mapped to continuous valence–arousal targets.

The transformer paradigm has introduced another major shift in MER. Transformer-based models are particularly effective at capturing long-range dependencies and global context, enabling not only song-level classification but also the modeling of emotional trajectories across time. Several studies have demonstrated that transformer encoders with self-attention pooling outperform traditional architectures in predicting valence and arousal [18, 19], while others focus on transitions and turning points in the emotional trajectory, introducing evaluation protocols such as windowed concordance correlation coefficients that emphasize temporal unfolding rather than static accuracy [18]. These contributions highlight the increasing importance of temporal modeling and sequence-level evaluation in MER. A related architectural development is the Conformer [20], originally proposed for automatic speech recognition, which augments the Transformer with depthwise-separable convolutions to combine the local-pattern bias of CNNs with the global self-attention mechanism. Although introduced for speech, the Conformer's hybrid local-global structure makes it applicable to continuous audio regression tasks more broadly, including music emotion recognition.

A complementary line of research has emerged from the broader field of audio representation learning, where self-supervised pre-training on large unlabelled corpora has produced general-purpose encoders that can be transferred to downstream tasks with minimal supervision. wav2vec 2.0 [21] introduced a contrastive masked-prediction objective on raw waveforms, learning latent speech representations that subsequently transfer to a wide range of speech-related tasks. HuBERT [22] refined this paradigm by replacing the contrastive objective with a clustering-based discrete-target prediction that encourages the encoder to capture phonetic and acoustic regularities at multiple temporal scales. While both models are trained on speech, their hidden representations have been shown to carry information relevant to non-speech audio classification, including affective tasks. Music-specific pre-training has been pursued in MERT [23], which adapts the masked-prediction paradigm to a large music corpus and incorporates auxiliary objectives based on acoustic and pitch-related targets, yielding representations that capture both timbral and harmonic structure. The use of such pre-trained encoders as feature extractors for music

emotion recognition is a natural extension of the deep-learning trajectory described above, and these models constitute the SOTA baselines against which the framework proposed in this work is compared in Sect. 9.

In addition to purely audio-based methods, multimodal approaches have been shown to offer substantial advantages. The integration of lyrics with acoustic signals through hierarchical or attention-based fusion consistently improves classification performance [24]. Recent work by Talaghat et al. [25] further demonstrates the effectiveness of multimodal integration, combining convolutional and recurrent layers with attention mechanisms to process audio features from the chorus and lyrics embeddings, introducing a new dataset of over 9,000 tracks with valence–arousal annotations. Real-time deployment scenarios have also received attention, with Hao et al. [26] proposing a multimodal fusion architecture that combines audio and visual streams under latency constraints, illustrating how the multimodal paradigm extends beyond accuracy improvements to address operational requirements relevant for interactive applications. Metadata such as language, demographics, or even self-reported mood have also been incorporated to capture variability in emotion perception [8, 10]. On the generative side, models that produce melodies under explicit emotional constraints [27] not only support applications in composition and interactive systems but also provide insights into which musical properties causally contribute to the elicitation of particular emotions.

A growing research frontier concerns personalization and explainability. Since emotional responses to music are highly subjective, approaches that adapt predictions to the individual listener have gained attention [6, 28]. These efforts rely on datasets that provide annotator profiles or contextual data, and they aim to build systems that are more accurate and user-centered. At the same time, explainable MER has emerged as an important theme, with methods designed to trace predictions back to sound sources or to mid-level perceptual features such as harmonic complexity, tonal stability, or rhythmic regularity [29, 30]. These studies not only improve the interpretability of models but also enhance scientific understanding of the mechanisms linking musical structure to emotional response.

Beyond recognition itself, several works explore how MER can be used in applications and interfaces. Some contributions propose visual mappings of musical emotions into intuitive modalities such as color or graphical displays [31, 32], while others integrate emotion recognition into adaptive music players that account for the listener's current state, for example by combining musical classification with facial expression recognition [33]. Such applications demonstrate the practical potential of MER in consumer technologies, music therapy, and human-computer interaction.

Finally, unification efforts aim to bridge the gap between categorical and dimensional paradigms. Multi-task frameworks that jointly learn discrete emotions and continuous valence–arousal values [34] show that both perspectives can be effectively integrated, benefiting from shared representations. Surveys and meta-analyses [2, 10, 35] underline that despite these advances, important challenges remain. Chief among them are the inherent subjectivity and noise in annotations, evaluator variability, the limited generalization of models across datasets, and the need for evaluation metrics that adequately capture the dynamic nature of music-induced emotions. These open issues highlight the need for novel approaches capable of addressing both the technical and conceptual limitations of current systems.

While the existing literature has made substantial progress in advancing Music Emotion Recognition through architectural innovations, multimodal fusion, and increasingly sophisticated deep learning models, comparatively less attention has been devoted to the systematic investigation of how different families of audio descriptors compare in their ability to predict continuous emotional trajectories, and how their complementary strengths can be leveraged through principled ensemble strategies. Many studies employ a single descriptor family or compare models within a fixed feature space, leaving open the question of how the choice of audio representation interacts with model architecture and learning paradigm. Similarly, while ensemble learning has been explored in discrete emotion classification, its potential for combining heterogeneous descriptor-model combinations in continuous valence-arousal prediction remains underexplored.

The present work addresses these gaps by providing a comprehensive and methodologically rigorous comparison of three theoretically diverse audio descriptor families, evaluating them in combination with six machine

learning paradigms and integrating their predictions through a multi-level ensemble framework that operates at the decision, feature, and hybrid levels. The framework is benchmarked against pre-trained audio Transformers and validated with paired statistical tests, with explicit attention to interpretability through meta-learner weight analysis. The detailed components of this framework, including the descriptor families, the machine learning paradigms, the ensemble strategies, and the experimental protocol, are presented in the following sections.

3 DEAM dataset

The Database for Emotional Analysis of Music (DEAM) [4] is one of the most widely adopted resources for the study of continuous emotion recognition in music. It comprises 1,802 audio excerpts drawn from the Free Music Archive and Jamendo, covering a broad spectrum of genres such as rock, pop, electronic, classical, and ambient music. Each excerpt has a duration of 45 seconds, which offers sufficient temporal context for analyzing emotional dynamics while maintaining manageable computational requirements.

Emotional annotations in DEAM are grounded in Russell's circumplex model of affect [3], representing affective experience along two continuous dimensions: valence (from negative to positive affect) and arousal (from calm to excited states) (see Fig. 1). The annotations were collected through a large-scale crowdsourcing campaign, in which listeners continuously adjusted a joystick interface to indicate their perceived emotional state while listening to each audio clip. This setup allowed the capture of the temporal evolution of emotions, rather than static categorical judgments. Each musical excerpt was rated by multiple participants, with a median of nine raters per song, and the final reference trajectories were obtained by averaging their annotations, thereby reducing the influence of individual biases and idiosyncratic responses. Both valence and arousal values are normalized to the range $[-1, +1]$, where -1 denotes negative or calm emotions and $+1$ corresponds to highly positive or energetic states.

The dataset provides a relatively balanced coverage of the valence–arousal space, though certain biases are evident and consistent with affective patterns found in publicly available music. The valence distribution shows a mild positive skew (mean ≈ 0.15, $\sigma \approx 0.35$), reflecting the general prevalence of pleasant or uplifting content. The arousal dimension spans a broader range (mean ≈ 0.05, $\sigma \approx 0.42$), encompassing both calm and energetic excerpts. When discretized into the four quadrants of Russell's model, approximately 28% of the excerpts fall

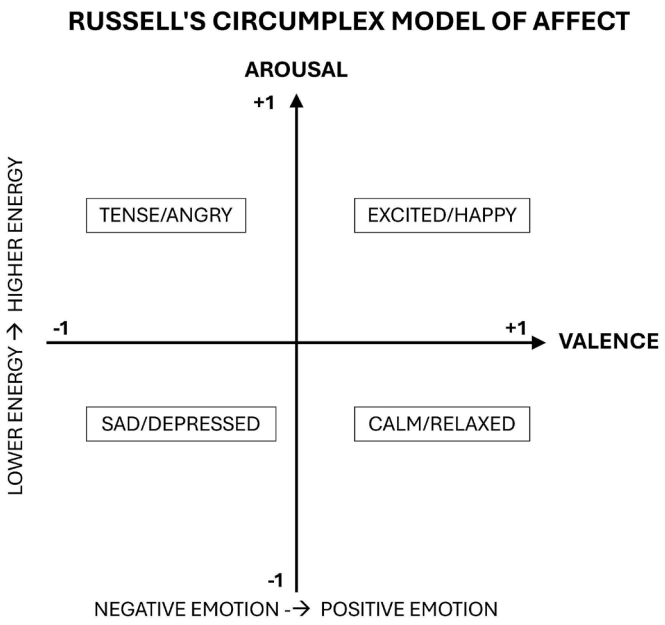


Fig. 1 Russell's circumplex model of affect. Emotions are organized in a two-dimensional space defined by valence (positive–negative) and arousal (high–low)

into the high-arousal/positive-valence quadrant, 30% into low-arousal/positive-valence, 22% into low-arousal/negative-valence, and 20% into high-arousal/negative-valence. This distribution indicates good overall coverage of emotional states, though slightly favoring positive valence.

Inter-rater reliability, as measured by Krippendorff's alpha, averaged 0.21 for valence and 0.19 for arousal, values that, though modest, are consistent with the high subjectivity of affective perception and with other datasets based on continuous emotion annotations. The aggregation of multiple raters mitigates this variability and provides a more stable estimate of perceived emotion trajectories. DEAM's annotation protocol also includes a 15-second silent "warm-up" period preceding each musical excerpt, intended to allow participants to become familiar with the joystick interface before the onset of the stimulus. Emotion annotations begin at the 15-second mark, aligning data collection with the active listening phase, and they are sampled every 0.5 seconds.

From a broader perspective, DEAM's design has several strengths that have established it as a benchmark dataset in affective computing. Its use of continuous, time-aligned annotations allows for fine-grained modeling of emotional variation across time, distinguishing it from corpora that rely solely on static or categorical labels. The inclusion of diverse musical genres supports cross-style generalization, while its open-access availability has made it a reference for reproducible experimentation and comparison across research groups.

Nevertheless, some limitations should be noted. The 45-second excerpts, while practical for annotation, may not fully capture the long-term emotional evolution found in complete musical works. The dataset is composed mainly of Western music, which may constrain cross-cultural generalization, and the subjective nature of emotion perception implies that even averaged ratings reflect collective tendencies rather than objective truth. In addition, a degree of genre imbalance is present, as popular or mainstream genres are more represented than niche or non-Western forms.

Despite these caveats, DEAM remains the most comprehensive and widely used benchmark for dynamic music emotion recognition. Its scale, annotation quality, and methodological rigor have established it as the foundation for evaluating predictive models, studying temporal affective patterns, and advancing multimodal approaches to the computational analysis of musical emotions. Table 1 summarizes the main characteristics of the DEAM dataset.

4 Methodology

The methodology presented here is designed to address these gaps through a comprehensive and empirically grounded investigation that systematically examines the interplay between audio descriptors, machine learning methods, and ensemble strategies. The goal is not to propose a novel architecture, but rather to provide evidence-based insights into best practices for building effective and interpretable MER systems using established and reproducible techniques. Six central research questions guide the methodological framework:

1. How do different families of audio descriptors, capturing spectro-temporal, psychoacoustic, and timbral information, compare in their ability to predict continuous valence and arousal?

Table 1 Summary of dataset characteristics and annotation statistics

Section	Statistics summary
Total excerpts	1,802 songs
Duration per excerpt	45 seconds
Frame duration	0.5 seconds
Total annotated frames	∼162,000
Annotation dimensions	Valence and Arousal (continuous, $[-1, +1]$)
Annotation method	Continuous joystick-based ratings
Median raters per song	9
Inter-rater agreement (α)	0.21 (valence), 0.19 (arousal)

2. Among widely used machine learning approaches, which modeling paradigms best capture the complex non-linear relationships between acoustic features and perceived emotion?
3. Can ensemble methods that integrate multiple descriptor–model combinations achieve meaningful improvements over individual systems, and which ensemble strategies prove most effective?
4. How do the three canonical levels of information fusion, namely decision-level (late), feature-level (early), and hybrid fusion, compare in this setting, and does combining them in a hybrid stacking meta-learner provide additional gains over either level taken in isolation?
5. Which descriptor families, learning paradigms, and individual features carry the predictive content of the framework, and how does this content distribute across the valence and arousal dimensions?
6. How does the proposed framework compare against state-of-the-art audio architectures, including both pre-trained Transformers used as feature extractors and a Conformer trained from scratch, and what does this comparison reveal about the trade-off between accuracy, interpretability, computational cost, and reliance on external data?

The methodological workflow follows a hierarchical experimental design articulated across three stages (see Fig. 2).

In the first stage, an exhaustive comparison is conducted across three distinct families of audio descriptors combined with six complementary machine learning methods, representing diverse modeling paradigms. The chosen descriptors represent fundamentally different philosophies of audio characterization: OpenSMILE, which provides a comprehensive signal-processing-based representation; Psychoacoustic Annoyance (PAA) features, which model auditory perception according to psychoacoustic principles; and Librosa descriptors, which include the standard timbral and rhythmic features commonly adopted in music information retrieval. The six machine learning methods, namely Long Short-Term Memory (LSTM) networks, Multi-Layer Perceptrons (MLP), Random Forests (RF), TabNet, Partial Least Squares Regression (PLSR), and Extreme Gradient Boosting (XGBoost), were selected to encompass diverse learning paradigms that capture temporal dynamics, non-linear mappings, and ensemble-based relationships, respectively.

In the second stage, three canonical fusion levels are systematically evaluated to exploit complementarity among base models. Decision-level (late) fusion combines the predictions of the base models through three strategies of increasing sophistication: arithmetic mean averaging, providing an unweighted baseline; Leave-One-Fold-Out (LOFO) weighted averaging, learning optimal linear weights through validation-based optimization;

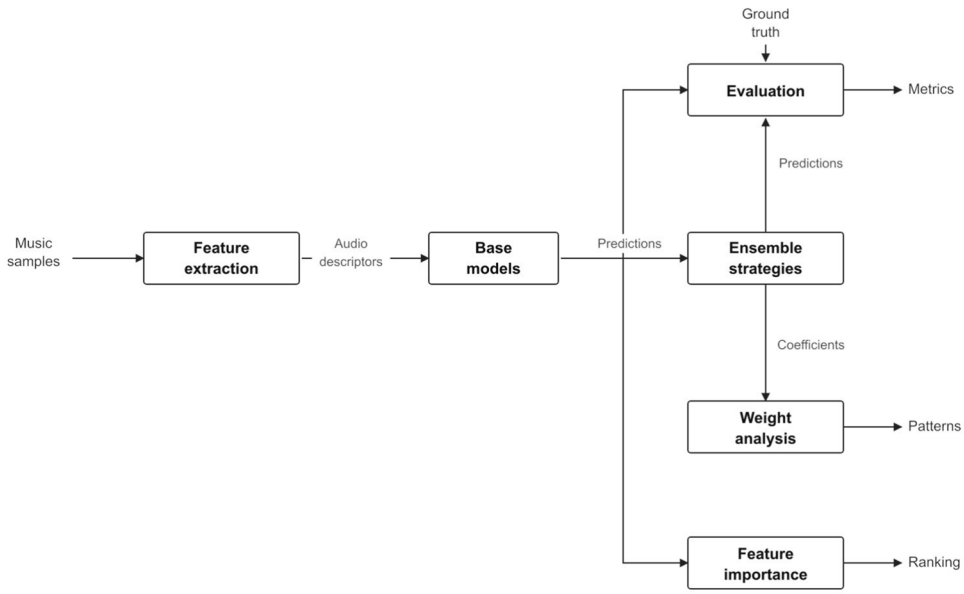


Fig. 2 Overall workflow of the proposed framework

and stacking with Ridge regression meta-learning, enabling dimension-specific combination of predictions. Multiple late-fusion configurations are evaluated, ranging from same-method ensembles that combine identical paradigms across descriptors to cross-method ensembles that mix diverse approaches across feature representations. Feature-level (early) fusion concatenates the three descriptor families into a single representation on which one base model per learning paradigm is trained. Hybrid fusion combines the predictions of the early-fusion and late-fusion base models through a Ridge stacking meta-learner. The three fusion levels are evaluated under the same cross-validation protocol, allowing a direct comparison of their relative effectiveness.

In the third stage, detailed analytical procedures are applied to extract insights from the ensemble results, to position the framework against the state of the art, and to validate the observed gains. These analyses include the interpretation of meta-learner weights to assess the relative contribution of each model and to expose dimension-specific contribution patterns; the comparison against four state-of-the-art audio architectures, namely three pre-trained Transformers (MERT, HuBERT, wav2vec 2.0) evaluated under a unified multi-layer weighted-sum probing protocol and a Conformer trained from scratch on the dataset's mel-spectrograms; the application of paired Wilcoxon signed-rank tests on the per-fold prediction errors to assess the statistical significance of the gains achieved by the ensemble strategies over the best individual base models; and a per-feature permutation-importance analysis of the three best XGBoost base models, one per descriptor family, to identify which specific features within each descriptor pool carry the predictive content. The objective is not only to quantify improvements in prediction accuracy but also to understand why particular combinations outperform others and to ground the reported improvements in formal statistical evidence, providing guidance for future methodological design in MER research.

The rationale underlying this methodological framework is grounded in four guiding principles.

First, diversity over novelty: rather than introducing architectural innovations, this work emphasizes the systematic exploration of diversity in descriptors, model types, and ensemble strategies. Theoretical and empirical results in ensemble learning [36] demonstrate that combining diverse models often yields greater improvements than refining a single homogeneous architecture.

Second, empirical rigor: all experimental components adhere to strict methodological controls to ensure fair comparison and replicability. The file-wise cross-validation protocol, for instance, avoids data leakage by ensuring that all frames from a given song appear exclusively in either the training or testing set, a crucial consideration in sequential audio analysis. Reported gains are further validated through paired Wilcoxon signed-rank tests on the per-fold prediction errors, providing formal statistical evidence that the observed improvements are not due to fold-level variability.

Third, interpretability and transparency: the methods chosen are well-established and analytically tractable, allowing insights into their internal mechanisms. The use of Ridge regression as a linear meta-learner makes it possible to examine the contribution of each base model and the structure of learned combinations across emotional dimensions.

Finally, practical applicability: by relying exclusively on publicly available libraries and standardized feature extraction tools (OpenSMILE, Librosa, and psychoacoustic models), the methodology ensures full reproducibility and adaptability for other research groups or application contexts.

In addition, this framework embraces the principle that the effectiveness of a music emotion recognition system depends not only on the sophistication of its modeling components but also on their complementarity and integration. Through rigorous evaluation and controlled ensemble design, the proposed methodology aims to provide a clear, interpretable, and generalizable understanding of how diverse feature spaces and modeling approaches jointly contribute to the prediction of emotional trajectories in music.

The following sections detail each methodological component in depth. The next section describes the extraction of audio descriptors and feature preprocessing procedures, followed by an overview of the machine learning methods adopted and their configurations, and finally the ensemble learning strategies developed to integrate model predictions into a unified, optimized framework.

5 Audio descriptors

The selection of appropriate audio descriptors is fundamental to the success of music emotion recognition systems, as these features represent the bridge between the raw acoustic signal and the higher-level emotional interpretation. In this study, three distinct families of audio descriptors were investigated, each representing a different theoretical and methodological approach to the characterization of musical audio. The chosen descriptor families, namely OpenSMILE, Psychoacoustic Annoyance (PAA), and Librosa, were selected to maximize diversity in feature representation while maintaining empirical and theoretical validity within music information retrieval and affective audio analysis.

To ensure fair comparison across descriptor families and numerical stability during model training, all feature sets undergo consistent preprocessing including standardization, outlier mitigation, and imputation of missing values. The complete preprocessing protocol is detailed in Sect. 8 (Experimental Setup).

5.1 OpenSMILE: comprehensive spectro-temporal features

OpenSMILE (Open-Source Speech and Music Interpretation by Large-Space Extraction) [37, 38] is one of the most widely adopted frameworks for the systematic extraction of acoustic features. Originally developed for speech emotion recognition, OpenSMILE has been extensively validated across various audio domains, including music analysis, and is considered a benchmark tool for standardized feature computation. It provides large, well-documented feature sets that capture a wide spectrum of acoustic phenomena through hierarchical signal processing.

For this study, we employed the Audeering implementation available at the GitHub repository [39]. The extraction was performed on non-overlapping 0.5-second frames, in line with the temporal resolution of the DEAM annotations. For each frame, OpenSMILE extracts a comprehensive set of 988 acoustic features encompassing spectral, prosodic, and voice quality descriptors. These features include spectral descriptors such as MFCCs (Mel-Frequency Cepstral Coefficients), spectral centroid, rolloff, flux, and energy in multiple frequency bands; prosodic features including fundamental frequency (F0), jitter, shimmer, and their statistical functionals; voice quality measures such as Harmonics-to-Noise Ratio (HNR) and spectral harmonicity; and energy-related features including RMS energy, loudness, and their derivatives.

The full 988-dimensional feature pool produced by the emobase_v2 configuration is passed to the downstream models without further reduction. Since emobase_v2 is itself a curated affective-computing set, in which low-level descriptors and statistical functionals are already selected for their relevance to emotional content, applying an additional supervised feature selection would introduce a further dataset-specific filtering of an already filtered representation. The complete vector is therefore retained, and the per-feature permutation-importance analysis described in Sect. 9 identifies a posteriori which descriptors actually carry the predictive content.

OpenSMILE features embody a theory-driven and comprehensive approach to audio description. The framework has been designed based on empirical evidence of feature relevance to affective communication and validated across diverse emotion recognition contexts. Its inclusion in this work ensures a rich, multi-faceted representation of the acoustic signal grounded in well-established acoustic correlates of emotion.

5.2 PAA: psychoacoustic annoyance features

The Psychoacoustic Annoyance (PAA) model [40, 41] represents a fundamentally different approach to audio characterization. Rather than relying on purely signal-processing criteria, PAA features are grounded in perceptual psychology and auditory physiology, modeling how humans actually experience sound. Originally developed for the quantification of acoustic discomfort, the PAA framework has since been adapted for music analysis, particularly in studies investigating the perceptual correlates of emotional arousal and valence.

The PAA pipeline computes, for each 0.5-second frame, the five core perceptual quantities prescribed by the Zwicker–Fastl framework [40], namely Loudness, Sharpness, Roughness, Fluctuation Strength, and Tonality. The extraction is performed with MoSQITo [42], a Python implementation of the Zwicker–Fastl psychoacoustic models. Loudness is computed in sones, accounting for frequency-dependent auditory sensitivity and temporal integration. Sharpness, expressed in acum units, quantifies the perceived high-frequency energy and relates to the brightness or shrillness of sound. Roughness, measured in asper units, captures rapid amplitude modulations in the 15–300 Hz range that produce sensations of tension or harshness, often associated with high arousal. Fluctuation strength, measured in vacil units, quantifies slower modulations (0.5–20 Hz) corresponding to rhythmic pulsations and beats. Tonality represents the degree of harmonicity versus noisiness, reflecting the distinction between tonal and atonal content.

In addition to these five descriptors, statistical functionals (mean, variance, and peak values) are computed to summarize temporal evolution within each frame, yielding a total of 20 features per frame. These descriptors are designed to reflect perceptual mechanisms such as critical band masking, frequency-dependent loudness perception, and temporal integration, which collectively define how sound energy is processed by the human auditory system. The resulting features thus bridge the gap between physical sound properties and perceptual experience.

PAA's inclusion in this study provides a perception-centered perspective that complements the more physically grounded OpenSMILE features. The explicit modeling of auditory phenomena such as roughness and sharpness is particularly relevant to emotion recognition, as previous psychological research [1] has linked these perceptual attributes to arousal and valence dimensions, respectively. By incorporating psychoacoustic modeling, PAA captures aspects of musical affect that are otherwise invisible to traditional signal-based descriptors, offering a valuable contribution to the diversity of feature representations explored.

5.3 Librosa: standard MIR timbral and rhythmic features

Librosa [43] is a well-established Python library for audio and music analysis that provides a standard set of features commonly used in Music Information Retrieval (MIR) research. While less specialized than OpenSMILE or PAA, Librosa offers an efficient and flexible implementation of the most frequently used timbral, spectral, and rhythmic descriptors. Its inclusion serves as a baseline aligned with conventional MIR practice and allows for direct comparison between standard descriptors and more specialized or perceptually motivated representations.

From each 0.5-second frame, a combination of spectral, timbral, rhythmic, and energy-related features is extracted. Spectral features include the spectral centroid (representing the center of mass of the spectrum and correlating with perceived brightness), spectral bandwidth (indicating spectral spread), spectral contrast (capturing energy differences between peaks and valleys across sub-bands), spectral roll-off (the frequency below which 85% of spectral energy is contained), and spectral flatness (measuring the degree to which the spectrum is noise-like). Timbral features are represented by 13 Mel-Frequency Cepstral Coefficients (MFCCs), which describe the spectral envelope on a perceptually motivated mel scale and are widely used in both speech and music emotion recognition. Additional harmonic information is captured by Chroma features, which represent the relative distribution of pitch classes across octaves. Rhythmic features include Tempo, estimated in beats per minute (BPM) via autocorrelation of the onset strength envelope, and the Zero-Crossing Rate (ZCR), which reflects the rate of signal polarity changes and correlates with percussiveness or noisiness. Finally, the Root Mean Square (RMS) energy provides a measure of overall loudness.

Time-varying features such as spectral descriptors, MFCCs, and chroma vectors are statistically summarized over the 0.5-second window through their mean, standard deviation, and range, resulting in a total of 77 features per frame. All features are computed using a Short-Time Fourier Transform (STFT) with a 2048-sample window and 512-sample hop size at 44.1 kHz sampling rate. Mel-scale computations use 128 mel bands spanning 0–8000 Hz, and chroma features are extracted using the Constant-Q Transform (CQT) to enhance pitch resolution.

Librosa features represent a pragmatic and domain-standard approach to MIR-based music analysis. Their inclusion allows benchmarking against widely adopted descriptors and supports interpretability, as many of these

features have well-established perceptual and musical meanings (for example, chroma for harmonic structure and tempo for rhythmic energy). The Librosa set thus provides a balanced complement to the theory-driven OpenSMILE and perception-oriented PAA descriptors.

5.4 Rationale for descriptor selection and diversity

The selection of the three descriptor families was guided by the objective of maximizing diversity across theoretical foundations, granularity, domain specificity, and computational philosophy. The feature sets differ markedly in dimensionality, with OpenSMILE being high-dimensional (988 features, used in full), Librosa moderately dimensional (77 features), and PAA low-dimensional (20 features). They also differ in domain orientation, with OpenSMILE being general-purpose across speech and music, PAA focusing on auditory perception modeling, and Librosa specifically tailored for music analysis. These variations yield distinct inductive biases for learning algorithms and are the structural precondition that makes ensemble learning meaningful in this setting. If the three families captured largely overlapping information, combining them would yield diminishing returns; if they capture complementary information, their integration through ensemble strategies can compensate for weaknesses in individual feature spaces. The expectation is also dimension-specific: descriptors emphasizing loudness and roughness (PAA) may correlate strongly with arousal, while harmonic and chromatic descriptors (Librosa) may prove more predictive for valence. Whether and to what extent these theoretical expectations are reflected in the actual contribution of each family to the trained models is one of the empirical questions addressed in Sect. 9.

6 Machine learning methods

The selection of suitable machine learning methods is central to effectively capturing the complex and non-linear relationships that link acoustic features to perceived emotional dynamics. In this study, a diverse set of algorithms was investigated, encompassing both traditional and modern paradigms of supervised regression. Each method was implemented to predict continuous valence and arousal trajectories jointly within a unified multi-output framework, ensuring consistency across modeling approaches and enabling direct comparability. This design allows subsequent analysis to determine, empirically, which families of models provide the best balance between predictive accuracy, interpretability, and computational efficiency.

The investigated algorithms include Long Short-Term Memory (LSTM) networks, Multi-Layer Perceptrons (MLP), Random Forests (RF), TabNet, Partial Least Squares Regression (PLSR), and Extreme Gradient Boosting (XGBoost). Together, these methods span a wide methodological spectrum, from deep neural architectures capable of temporal modeling to classical linear and ensemble-based techniques optimized for tabular data. Each contributes a distinct inductive bias, providing complementary perspectives on the mapping between acoustic features and emotional dimensions.

6.1 Long Short-Term Memory networks (LSTM)

Long Short-Term Memory (LSTM) networks [44] constitute a specialized form of Recurrent Neural Network (RNN) explicitly designed to capture long-range temporal dependencies in sequential data. Their internal gating mechanisms allow selective retention and forgetting of information over time, mitigating the vanishing gradient problem typical of conventional RNNs. This makes them particularly well suited for modeling emotional trajectories in music, where affective states evolve continuously and are influenced by preceding acoustic context.

The LSTM models used in this study operate directly on temporal sequences of features extracted at 0.5-second intervals, enabling the network to learn temporal patterns that correspond to gradual emotional transitions. Two recurrent layers are stacked to capture both low and high level temporal dependencies, followed by a fully connected output layer with two linear units representing valence and arousal. The model jointly minimizes the

mean squared error across both dimensions, promoting the emergence of shared internal representations that encode the acoustic correlates of musical affect. By design, the LSTM's ability to maintain temporal memory provides an explicit mechanism for tracking the evolving emotional content of music, making it a conceptually appropriate choice for continuous affect prediction.

6.2 Multi-Layer Perceptron (MLP)

Multi-Layer Perceptrons (MLPs) [45] represent one of the most fundamental yet versatile forms of artificial neural networks, capable of learning non-linear mappings between high-dimensional input features and continuous output variables. Although MLPs do not model temporal dependencies explicitly, they are particularly effective when applied to frame-level descriptors that already incorporate statistical or contextual information, as is the case for the features used in this work.

The MLP architecture consists of multiple fully connected layers with non-linear activations, culminating in a two-unit linear output layer corresponding to valence and arousal. The network is trained using a Mean Squared Error (MSE) objective computed jointly over both emotional dimensions. Despite their structural simplicity, MLPs are powerful universal approximators and can efficiently capture complex interactions among acoustic descriptors. Their computational efficiency and straightforward training dynamics make them a valuable baseline for assessing the contribution of explicit temporal modeling in recurrent architectures.

6.3 Random Forest (RF)

Random Forests [46] are ensemble learning methods that construct a large number of decision trees, each trained on a random subset of the data and feature dimensions. The final prediction is obtained as the mean output of all trees, resulting in a model that is both robust and interpretable. In the context of continuous emotion prediction, Random Forests provide a non-parametric alternative to neural networks, capable of modeling highly non-linear relationships while remaining resistant to overfitting and insensitive to feature scaling.

In this study, Random Forest regressors were implemented in a multi-output configuration, predicting valence and arousal jointly through a single ensemble structure. Each split within the trees was optimized to minimize the joint squared error, consistent with the MSE objective, enabling the model to exploit shared dependencies and covariances between the two emotional dimensions. This design allows the forest to capture inter-dimensional relationships that are often ignored in independent regressors, while maintaining the interpretability typical of tree-based models. Feature importance analyses can be derived directly from the trained ensemble, offering insights into which acoustic parameters most strongly influence emotional perception.

6.4 TabNet

TabNet [47] represents a modern deep-learning approach specifically tailored to structured tabular data. Its architecture combines sequential attention mechanisms with instance-wise feature selection masks, allowing the model to focus dynamically on the most relevant features at each decision step. Unlike traditional feed-forward networks that treat all input dimensions uniformly, TabNet performs conditional computation, selectively attending to different subsets of features depending on the input instance. This enables a balance between interpretability and non-linear modeling capacity.

In the present framework, TabNet is applied to frame-level acoustic descriptors to learn complex mappings between features and continuous emotion values. By leveraging attention-driven feature selection, it can identify which acoustic cues contribute most strongly to each emotional dimension in a context-dependent manner. Although originally designed for general tabular data, TabNet's interpretability and adaptive feature selection make it a compelling candidate for music emotion recognition, where the relative importance of different descriptors (for example, spectral versus rhythmic) may vary over time and across emotional states. The network is

configured in a multi-output regression mode to predict valence and arousal jointly, sharing internal representations across both emotional dimensions.

6.5 Partial Least Squares Regression (PLSR)

Partial Least Squares Regression (PLSR) [48] serves as a classical, interpretable baseline within this comparative framework. PLSR projects high-dimensional input features onto a smaller set of latent components that maximize covariance with the target variables. In this work, PLSR operates in multi-target mode, predicting valence and arousal simultaneously from a shared set of latent components. This dual projection simultaneously reduces dimensionality and retains the directions of greatest predictive relevance, making PLSR particularly effective when features are highly correlated or when the feature space exceeds the number of training samples.

Although inherently linear, PLSR provides valuable insight into the degree of linear separability between acoustic features and emotional dimensions. By analyzing the weights and loadings of the latent components, it is possible to interpret which feature groups align most strongly with valence or arousal. Its inclusion ensures that the methodological framework spans the full continuum from simple, interpretable linear models to highly expressive non-linear learners.

6.6 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) [49] extends the family of tree-based ensemble methods through iterative boosting, where successive trees are trained to correct the residual errors of previous ones. Each iteration focuses learning capacity on the most difficult samples, gradually refining the prediction surface. XGBoost integrates regularization mechanisms through both L1 and L2 penalties, ensuring stability and preventing overfitting even in high-dimensional spaces.

Within this study, XGBoost was implemented in a multi-output configuration to jointly predict valence and arousal. By optimizing a shared objective function across both targets, the model is able to capture joint dependencies between emotional dimensions while benefiting from the high expressive power and efficiency of gradient-boosted ensembles. Its combination of non-linear flexibility, robustness, and interpretability makes it a natural extension of the Random Forest paradigm, capable of learning fine-grained relationships between acoustic cues and emotional responses.

6.7 Rationale for method selection and diversity

Each of the methods described above contributes a distinct perspective on the prediction of continuous emotional trajectories. Neural architectures such as LSTMs and MLPs emphasize representation learning and hierarchical abstraction, while tree-based and boosting methods focus on partitioning the feature space through interpretable decision structures. TabNet bridges these paradigms by introducing attention-based feature selection in a tabular context, and PLSR anchors the comparison within a well-understood linear framework. This methodological diversity is fundamental to the ensemble stage that follows, in which model outputs will be integrated through meta-learning to explore complementarity and redundancy among predictors. To keep the ensemble experimental design tractable while preserving methodological diversity, the composition of the ensemble will be guided by an empirical consistency criterion: only paradigms that perform competitively across all three descriptor families will be retained as base learners in the ensemble stage. The corresponding selection is detailed in Sect. 9, after the per-descriptor base-model results are presented.

7 Ensemble learning strategies

Ensemble learning represents a powerful paradigm in predictive modeling, based on the premise that combining multiple diverse models often yields superior generalization performance compared to any single constituent model. This advantage arises from the reduction of both variance and bias through aggregation, as different models capture complementary aspects of the data and their individual errors tend to average out. In the context of music emotion recognition, ensemble methods offer a particularly appealing solution to the inherent subjectivity and complexity of the task, where no single model or descriptor family can fully capture the multifaceted relationship between musical structure and perceived emotion.

The multi-classifier literature [50] distinguishes three canonical levels at which information from heterogeneous sources can be combined: decision-level (or late) fusion, where the predictions of independently trained base models are aggregated; feature-level (or early) fusion, where the input representations are merged before training a single model on the joint representation; and hybrid fusion, where predictions of early- and late-fusion models are combined through an additional meta-learning stage. We evaluate all three levels on the continuous valence–arousal regression task, under a unified 5-fold file-wise cross-validation protocol that allows a direct comparison of their relative effectiveness.

The ensemble framework developed in this study builds upon the foundations established by [36] and [51]. In all three decision-level strategies, the first stage consists of the late-fusion base learners, namely the descriptor–method combinations retained according to the consistency criterion introduced in Sect. 6.7 (the exact composition is reported in Sect. 9). Each base model is trained under the same 5-fold file-wise cross-validation protocol, producing out-of-fold predictions that serve as the foundation for the three combination strategies considered at the decision level: arithmetic mean averaging, Leave-One-Fold-Out (LOFO) weighted averaging, and Ridge regression stacking, in order of increasing sophistication. Each strategy is described in turn below.

In the arithmetic mean approach, ensemble predictions are computed as the simple average of base model predictions for each emotional dimension. Formally, for a set of N base models producing predictions $\hat{y}_1$, $\hat{y}_2$,..., $\hat{y}_N$, the ensemble prediction is given by $\hat{y}_{\text{ensemble}} = (1/N)\ \Sigma_i \hat{y}_i$. This method requires no training or parameter optimization and serves as a baseline to quantify the benefit of model diversity alone, independent of any learned weighting scheme.

In the LOFO weighted averaging approach, optimal linear weights are learned for each base model through a leave-one-fold-out optimization procedure. For each cross-validation fold *k*, weights $\mathrm{w} = [w_1, w_2, \ldots, w_N]$ are optimized on the out-of-fold predictions from the remaining folds, minimizing the Mean Euclidean Distance (MED) in the valence–arousal space, defined as the average per-frame Euclidean distance between predicted and reference points. The learned weights are then applied to fold k to generate ensemble predictions. This process is repeated for all folds, ensuring that weight optimization never sees the test data. The weights are shared across both valence and arousal dimensions, providing a global weighting scheme that balances performance across both emotional axes.

In the stacking approach, a Ridge Regression meta-learner is trained to combine base model predictions. The meta-learner receives as input the predicted valence and arousal values from each of the base models and learns to produce optimized final predictions. Ridge regression was chosen for its interpretability, simplicity, and robustness against collinearity among predictors. It minimizes the mean squared error while imposing an L2 penalty on the magnitude of the regression coefficients, thereby preventing overfitting and ensuring stable weight estimation even when the base models are highly correlated. The regularization coefficient ($\alpha = 1.0$) was selected empirically, testing different values, to balance bias and variance while maintaining interpretability of the resulting weights. The Ridge regressor performs multivariate regression, jointly predicting both valence and arousal from the concatenated base model predictions. This allows the meta-learner to capture any interdependencies between the emotional dimensions while learning the optimal combination of base models.

A key methodological refinement in the stacking approach is the use of a Leave-One-Fold-Out training scheme, designed to preserve complete independence between the base and meta-learning stages. The meta-learner is trained exclusively on the out-of-fold predictions generated during cross-validation, ensuring that the meta-model never sees predictions from base learners on the same data used for their training. This strategy effectively eliminates information leakage and produces a truly generalizable ensemble model. Once trained, the meta-learner is applied to the test predictions from each fold to obtain the final ensemble outputs for valence and arousal.

The three strategies just described all operate at the decision level, combining the outputs of independently trained base models. The remainder of this section turns to the other two fusion levels introduced above, in which integration occurs at the input space (early fusion) or across both levels (hybrid fusion).

In feature-level (early) fusion, the three descriptor families are concatenated frame by frame into a single 1085-dimensional joint representation (988 OpenSMILE + 77 Librosa + 20 PAA), and one base model per retained learning paradigm is trained on the resulting feature space, ensuring a fair comparison with the late-fusion ensemble under identical learning capacity. Cross-descriptor interactions, which decision-level fusion can only recover indirectly through the meta-learner, are exposed directly to the early-fusion model, which is free to allocate capacity across descriptor families as the learning objective dictates. While Bejani et al. [52] include an ANOVA-based feature-ranking step on the joint representation, we pass the full concatenation to the early-fusion models without dimensionality reduction, consistently with the no-selection policy adopted on the OpenSMILE side. The resulting out-of-fold predictions are directly comparable, fold by fold, with their late-fusion counterparts.

Hybrid fusion combines the predictions of the early- and late-fusion base models through a Ridge stacking meta-learner identical in form and training scheme to the decision-level one, with the input pool enlarged to include both fusion levels. This design lets the meta-learner re-weight the contributions of decision-level and feature-level information in a dimension-specific manner, allocating different mixtures to valence and to arousal. Since only the input pool grows, the hybrid stage adds negligible training cost once the base models are available.

The rationale for evaluating multiple combination mechanisms operates at two complementary levels. Within the decision-level family, the three strategies span a simplicity–adaptability trade-off: arithmetic mean averaging offers maximum simplicity and interpretability but may be suboptimal when base models have substantially different levels of accuracy or when their errors are correlated; LOFO optimization addresses this limitation by learning data-driven weights, but it remains constrained to a single set of global weights applied uniformly across both emotional dimensions; stacking with meta-learning relaxes this constraint by allowing dimension-specific weighting and the potential incorporation of nonlinear interactions. Across fusion levels, the comparison addresses a different question, namely whether the locus of integration matters: decision-level fusion preserves the inductive biases of the individual base learners and combines only their outputs; feature-level fusion lets a single model exploit cross-descriptor interactions directly in the input space; hybrid fusion attempts to retain both advantages by stacking the two paradigms. By comparing the three fusion levels and the three decision-level strategies jointly, the study aims to quantify the performance gains associated with each layer of the integration hierarchy in the context of continuous music emotion recognition.

A common feature of all combination strategies considered is that they remain analytically tractable. The Ridge meta-learner, in particular, exposes a set of linear coefficients whose magnitude and sign can be directly inspected, both across base models in the decision-level setting and across fusion levels in the hybrid setting. Because the meta-learner is fitted separately on valence and on arousal, these coefficients are naturally dimension-specific: a base model that contributes strongly to one emotional dimension is not forced to contribute equally to the other, and the resulting weight profile can be read as a quantitative summary of which descriptor–method combinations carry the predictive content for each dimension. This interpretability is a methodological advantage over deep neural ensemble models that function as opaque combiners, and it is exploited in Section 9 to characterize the relative contribution of descriptor families, learning paradigms, and fusion levels to the prediction of valence and arousal.

8 Experimental setup

The experimental evaluation was designed to ensure methodological rigor, reproducibility, and fair comparison across all model–descriptor combinations. A uniform cross-validation protocol was applied to all base models, with consistent preprocessing and evaluation metrics. Particular attention was devoted to preventing data leakage and ensuring that all models were trained and evaluated under identical conditions.

8.1 Cross-validation protocol and preprocessing

To ensure robust and generalizable performance estimates, a 5-fold cross-validation strategy was adopted for all experiments. Critically, the splitting was performed at the file level rather than the frame level, meaning that all 0.5-second frames extracted from a given musical excerpt were assigned exclusively to either the training set or the test set within each fold, but never to both. This file-wise splitting is essential to prevent data leakage, as frames from the same song are highly correlated and would otherwise allow models to exploit temporal dependencies across the train-test boundary. The dataset's 1,802 audio files were divided into five approximately equal subsets, with each fold using four subsets for training (approximately 1,442 files) and one for testing (approximately 360 files). The same fold assignments were used consistently across all base models, ensuring that differences in performance reflect genuine differences in modeling capacity rather than variations in training data composition.

All audio files were segmented into non-overlapping 0.5-second frames, aligned with the temporal resolution of the DEAM emotion annotations. This uniform segmentation was applied consistently across all three descriptor families (OpenSMILE, PAA, and Librosa), ensuring temporal synchronization and enabling direct comparison and combination within ensemble configurations. The choice of frame-level prediction, rather than song-level aggregation, is motivated by two considerations. First, DEAM provides continuous annotations at 0.5-second resolution, and operating at this granularity fully exploits the available temporal information. Second, frame-level predictions preserve the ability to analyze emotional trajectories and temporal dynamics; song-level or segment-level metrics can be readily derived through appropriate aggregation (for example, mean or median) when required by specific applications, whereas the reverse, namely recovering fine-grained dynamics from aggregated predictions, is not possible.

For each frame, audio descriptors were extracted using the respective toolkits, and the resulting feature matrices were subjected to a uniform preprocessing pipeline applied consistently across all descriptor families. To ensure numerical stability and fair comparison across feature spaces with different scales, all features were standardized to zero mean and unit variance through z-score normalization, where the mean and standard deviation were computed exclusively from the training set within each cross-validation fold. The same normalization parameters were then applied to the corresponding test set, ensuring that no test-set statistics influenced the preprocessing. To mitigate the influence of extreme outliers that may arise from atypical audio content or extraction artifacts, feature values were clipped to the range $[\mu - 5\sigma, \mu + 5\sigma]$ after standardization, where μ and σ refer to the training-set statistics. In cases where feature extraction yielded undefined or missing values (for example, missing fundamental frequency in unvoiced segments, or undefined tempo estimates in near-silent passages), these values were imputed using the mean value of the corresponding feature dimension, computed from the training set.

8.2 Hyperparameter configuration

Each machine-learning method was configured with a single fixed set of hyperparameter values, applied uniformly across all 5-fold cross-validation runs and across all descriptor families; the specific values used for each model are detailed below. For Random Forest, XGBoost, TabNet, and PLSR, library defaults were used as the starting point and lightly adapted to the data regime of the study; for LSTM and MLP, conventional configurations for sequential and tabular regression were adopted. No formal hyperparameter search was conducted. This

is a deliberate methodological choice, consistent with the goal of the study, which is the systematic comparison across descriptor families, learning paradigms, and fusion strategies rather than the pursuit of the best attainable performance for any individual model. Because the same configuration is used for every fold, every descriptor family, and every base learner of the same class, no model is given a tuning advantage over the others, and the comparisons reported in Sect. 9 reflect differences in modeling capacity under matched experimental conditions.

The LSTM architecture consisted of two stacked recurrent layers with 64 hidden units each, followed by a fully connected output layer with two units representing valence and arousal. Tanh activation was used for the LSTM cells, with linear activation for the output layer. Dropout regularization with a rate of 0.2 was applied after each LSTM layer to prevent overfitting. The network was trained using the Adam optimizer with a learning rate of 0.001, minimizing mean squared error computed jointly over both emotional dimensions. Training proceeded for up to 50 epochs with early stopping based on validation loss, using a patience of 10 epochs and a batch size of 128 frames.

The MLP architecture comprised three fully connected hidden layers with 128, 64, and 32 units respectively, followed by a two-unit output layer. ReLU activation was used for hidden layers, with linear activation for the output. Dropout with a rate of 0.3 was applied after each hidden layer. The network was trained using Adam with a learning rate of 0.001, a batch size of 256 frames, and up to 100 epochs with early stopping and a patience of 15 epochs.

The Random Forest regressor was configured with 300 trees, a maximum depth of 20, minimum samples per split of 10, and minimum samples per leaf of 5. At each split, a random subset of features equal to the square root of the total number of features was considered. The regressor operated in multi-output mode, with a single ensemble structure predicting both valence and arousal jointly, allowing the model to exploit shared dependencies between the two dimensions.

XGBoost was configured with 200 trees, a learning rate of 0.1, and a maximum depth of 6. Subsample and column subsample ratios were both set to 0.8 to introduce stochasticity and prevent overfitting, and L2 regularization with lambda equal to 1.0 was applied to the tree weights. The model was implemented in multi-output mode via a MultiOutputRegressor wrapper, optimizing a squared error objective for each dimension.

TabNet was configured with 5 decision steps, feature and output dimensions of 32 per step, and a relaxation factor of 1.5 to control the sparsity of attention masks. The network was trained with a batch size of 256, a learning rate of 0.02, and the Adam optimizer, for up to 100 epochs with early stopping and a patience of 20 epochs. A sparsity regularization term with lambda equal to 0.001 encouraged the model to focus on the most relevant features at each step. The output layer consisted of two units predicting valence and arousal simultaneously.

Partial Least Squares Regression was implemented with 20 latent components. The NIPALS algorithm was used for component extraction, and the model operated in multi-target mode, projecting the input features onto latent components that maximize covariance with both valence and arousal targets simultaneously. All features were pre-standardized as described above prior to PLSR training.

8.3 Evaluation metrics

Model performance was evaluated using a set of complementary metrics that capture different aspects of prediction accuracy. The primary multi-output metric is the Mean Euclidean Distance (MED), defined as the average over all frames of the Euclidean distance between predicted and ground-truth points in the two-dimensional valence–arousal plane:

$$MED = \frac{1}{N} \sum_{i} \sqrt{(\hat{v}_i - v_i)^2 + (\hat{a}_i - a_i)^2)} \tag{1}$$

where v_i and a_i denote the ground-truth valence and arousal values for frame i, and v_i and $\hat{a}_i$ their predicted counterparts, with N the total number of frames. The metric provides a holistic measure of accuracy in the

two-dimensional valence–arousal space, reflecting the perceptual distance between predicted and reference emotional trajectories. Lower values indicate better performance and more accurate reconstruction of emotional dynamics.

To complement the joint MED, two further error-based metrics are computed separately for valence and arousal. The Root Mean Squared Error (RMSE) along each dimension penalises larger errors more strongly through its quadratic dependence on the residuals:

$$RMSE_y = \sqrt{\frac{1}{N}\sum_i (\hat{y}_i - y_i)^2)} \tag{2}$$

The Mean Absolute Error (MAE) along each dimension provides a robust estimate of the average deviation in the same units as the target:

$$MAE_y = \frac{1}{N}\sum_i |\hat{y}_i - y_i| \tag{3}$$

In both definitions y stands for either valence or arousal, with the corresponding metric reported as RMSE_V, RMSE_A, MAE_V, and MAE_A in the rest of the manuscript.

Beyond error-based measures, the Concordance Correlation Coefficient (CCC) is computed to assess the agreement between predicted and ground-truth trajectories along each dimension:

$$CCC = (2\rho\sigma_{\hat{y}}\sigma_y)/(\sigma_{\hat{y}}^2 + \sigma_y^2 + (\mu_{\hat{y}} - \mu_y)^2) \tag{4}$$

where ρ denotes Pearson's correlation, σ the standard deviation, and μ the mean of predicted ($\hat{y}$) and ground-truth (y) values. Unlike MED, RMSE, and MAE, which measure absolute error magnitude, CCC captures how well predictions track the temporal variation of the emotional trajectory, penalising both poor correlation and systematic bias. CCC is reported separately for valence (CCC_V) and arousal (CCC_A) and is the standard metric in continuous affective computing benchmarks such as MediaEval. The full evaluation table thus reports, for every model and every ensemble configuration: the joint MED; the per-dimension RMSE_V, RMSE_A, MAE_V, MAE_A; and the per-dimension CCC_V, CCC_A.

All metrics are computed independently for each cross-validation fold, and final results are reported as mean values across the five folds.

8.4 Ensemble configurations

All ensemble configurations are built upon out-of-fold predictions generated under the same 5-fold file-wise cross-validation splits used for the base models, in order to preserve full independence between the base and meta-learning stages (the rationale for this scheme is detailed in Sect. 7). For LOFO weighted averaging, the weights are obtained by minimising the joint MED on the out-of-fold predictions of the training folds, then applied to the held-out fold. For Ridge stacking, the regularisation parameter α was selected from 0.1, 0.5, 1.0, 5.0, 10.0 through grid search on the training folds, with $\alpha = 1.0$ reported as the default configuration due to stable performance across all ensemble types.

For the early-fusion (feature-level) experiments, the three descriptor families are concatenated frame by frame into a single 1085-dimensional representation (988 OpenSMILE + 77 Librosa + 20 PAA features). The concatenation is performed after the per-family preprocessing described above. One base model per learning paradigm retained for the ensemble (see Sect. 9 for the retained set) is trained on the joint representation, with hyperparameter values identical to those used for the late-fusion base models of the same paradigm. The same 5-fold file-wise splits used for the late-fusion ensembles are reused, so that the per-fold predictions of the early-fusion models are directly comparable, fold by fold, with their late-fusion counterparts.

For the hybrid-fusion experiments, a Ridge regression meta-learner is trained on the concatenated pool of out-of-fold predictions produced by the late-fusion and early-fusion base models. The meta-learner is configured exactly as the one used in the decision-level Ridge stacking strategy (multi-output Ridge with $\alpha = 1.0$, fitted with non-negative coefficients and without intercept). The non-negativity constraint yields automatic sparsity in the resulting weight vector, providing direct visibility on how the meta-learner allocates contributions across both base models and fusion levels in a dimension-specific manner (analysis reported in Sect. 9).

8.5 SOTA architecture protocols

The four state-of-the-art audio architectures used as baselines, introduced in Sect. 2, are evaluated on the same 5-fold file-wise cross-validation splits as the proposed framework and with the same set of evaluation metrics, ensuring a strictly comparable benchmark. Two distinct protocols are required because the four architectures fall into two regimes: three are pre-trained encoders evaluated under a probing protocol, and the fourth is a Conformer trained from scratch on the DEAM corpus.

The three pre-trained Transformers, namely MERT [23], HuBERT [22], and wav2vec 2.0 [21], are evaluated in a frozen-encoder setting. For each model, the encoder weights are kept frozen and the hidden representations from all Transformer layers are aggregated via a learnable multi-layer weighted sum, where the layer-mixing weights are obtained as the softmax of a small set of trainable parameters. The aggregated representation is then mapped to valence and arousal predictions through a compact MLP probe head, trained jointly with the layer-mixing weights. Training uses the Adam optimizer with a learning rate of 1×10^{-3}, dropout 0.25, batch size 64, early stopping on a 10% file-wise held-out validation portion, and patience 15 epochs. The pre-trained encoder weights are accessed through the public model checkpoints released by the original authors, with no further fine-tuning of the encoder, so that the comparison reflects the quality of the pre-trained representations rather than that of any task-specific adaptation.

The Conformer [20], in contrast, is trained end-to-end on log-mel spectrograms (16 kHz, 25 ms window, 10 ms hop, 80 mel bands) extracted from the same annotated window. The architecture follows the canonical Conformer design with capacity reduced to match the scale of DEAM (2 blocks, encoder dimension 96, 4 attention heads, $\sim$ 700k parameters). Training uses AdamW with learning rate 1×10^{-3}, weight decay 1×10^{-4}, batch size 8, gradient clipping 1.0, and early stopping with patience 5 on a 10% file-wise held-out validation portion of each training fold; mel-spectrogram normalisation statistics are computed fold-internally to avoid leakage.

8.6 Per-feature importance analysis

Beyond the family-level comparison of base models, a per-feature importance analysis is run to identify which specific descriptors within each family carry the predictive content. The analysis is conducted with the permutation-importance procedure of Breiman [46] in the implementation pattern documented by scikit-learn. The procedure is model-agnostic and operates entirely on a held-out fold without retraining. For each feature *j* in turn, the values of that feature are randomly permuted across rows of the test set, leaving all other features untouched, and the model is queried on the perturbed test set. The deterioration of the MED metric is recorded and averaged over $n_{\text{repeats}} = 20$ independent permutations, yielding the importance score

$$I_j = \mathbb{E}\left[\text{MED}_j^{\text{shuffled}} - \text{MED}_0\right] \tag{5}$$

where MED_0 is the baseline MED measured on the unperturbed test set. I_j is expressed in the same units as the metric itself. It measures the absolute impact of feature *j* on the trained model: a large I_j indicates that the model relies heavily on feature *j*, while $I_j \approx 0$ indicates that the feature carries no predictive content. The score does not decompose the model's prediction into per-feature contributions, because permutation effects are not additive in the presence of correlated features and non-linear models. For ease of reading and to align with the standard

presentation in the permutation-importance literature, the raw score is also reported as the share of feature j within the family's total positive importance,

$$\text{share}_j = 100 \times \frac{I_j}{\sum_k I_k} \tag{6}$$

which is a 0–100 normalisation that sums to 100% across all features of the family and expresses how strongly a feature stands out within its descriptor set.

The analysis is performed on the three XGBoost base models of the (descriptor × method) grid, one per descriptor family. Adopting a single model class as the importance estimator across the three families serves three purposes. First, methodological consistency: it removes the model as a confounder in the cross-family comparison of importance scores, so that differences in the resulting rankings reflect properties of the features themselves rather than properties of different learning paradigms. Second, reproducibility: XGBoost training is fully deterministic under a fixed random seed and a fixed feature ordering, which allows the analysis to be reproduced bit-exactly from the predictions stored in the cross-validation output. Third, empirical competitiveness across families: as the base-model evaluation reported in Sect. 9.1 shows, XGBoost is the only learning paradigm that ranks consistently among the top performers on each of the three descriptor families, with stable mid-range MED values across OpenSMILE, PAA, and Librosa, whereas other paradigms either dominate on one family and collapse on the others or never reach the top tier. The importance scores are therefore computed on models whose predictive quality is empirically validated, and uniformly so, across the three feature spaces, rather than on a model class that happens to be a strong fit only for one of them.

To avoid multiplying the computational cost without a corresponding gain in precision, the importance is not computed fold-by-fold and then averaged. Instead, a single representative fold per family is selected as the fold whose per-fold MED is closest to the across-fold mean, the XGBoost model is retrained on the remaining four folds using the same hyperparameters and random seed as in Sect. 9.1, and the permutation loop is run on the held-out representative fold. The fold-selection rule is pre-specified, deterministic, and independent of the importance results, ensuring that the analysis reflects a typical fold rather than a best- or worst-case one. As an internal consistency check, the baseline MED computed by the retrained model on the representative fold is compared against the MED stored in the original cross-validation predictions to confirm that the model whose importance is being probed is the same model whose predictive performance is reported in Sect. 9.1. The number of permutations per feature ($n_{\text{repeats}} = 20$) is chosen so that the standard error of the mean importance, which scales as $\sigma/\sqrt{n_{\text{repeats}}}$ with σ the across-shuffle standard deviation, is small enough to separate signal from noise in a z-score sense, even on the high-dimensional OpenSMILE feature set. The results of the analysis are reported in Sect. 9.1.1.

8.7 Significance testing

To assess the statistical significance of the gains achieved by the ensemble strategies over the individual base models, paired Wilcoxon signed-rank tests are applied on the per-fold MED values, following the methodological recommendation of Demšar [53] for the non-parametric comparison of learning algorithms. Each test compares the per-fold MED of one base model with the per-fold MED of one ensemble configuration on the same five cross-validation folds, treating the two systems as matched pairs. The one-sided alternative "ensemble lower error than base" is adopted, which directly tests the directional hypothesis of interest. With n = 5 paired observations, the minimum attainable one-sided p-value is $1/2^5 = 0.031$ and the maximum attainable rank sum is W = 15; the simultaneous occurrence of W = 15 and p = 0.031 represents the strongest possible statistical signal that this experimental design can produce. The corresponding test outcomes are reported in Sect. 9.

8.8 Implementation

All experiments were implemented in Python 3.10 using standard libraries for machine learning and audio processing. Scikit-learn version 1.3.0 was used for Random Forest, Ridge Regression, evaluation metrics, and preprocessing utilities, while XGBoost version 2.0.0 provided the gradient boosting implementation. Deep learning models were implemented in PyTorch version 2.1.0, with TabNet using the PyTorch TabNet library version 4.0. Partial Least Squares Regression was implemented using scikit-learn's PLSRegression module. Audio feature extraction relied on OpenSMILE version 3.0 invoked via command-line interface, and Librosa version 0.10.0 for standard MIR descriptors. Psychoacoustic descriptors were extracted with MoSQITo [42]. Data manipulation was performed using NumPy version 1.24.0 and Pandas version 2.0.0. To ensure full reproducibility, all random number generators were seeded with a fixed value at the beginning of each experiment.

Throughout the experimental design, file-wise cross-validation, uniform preprocessing, fixed hyperparameter configurations, and the use of out-of-fold predictions for ensemble construction jointly ensure that the reported performance estimates are unbiased and comparable across all model–descriptor combinations and fusion strategies.

9 Experimental results

This section presents the experimental results obtained from the systematic evaluation of individual base models, fusion strategies, weight analysis, statistical testing, and state-of-the-art audio architecture baselines. The results are organised into four parts. Subsection 9.1 reports the performance of the eighteen descriptor–method combinations evaluated as base models, including a per-feature permutation-importance analysis, and identifies the four paradigms retained for the fusion stage according to the consistency criterion introduced in Sect.6.7. Subsection 9.2 evaluates the integration of the retained base models at the three fusion levels (late, early, hybrid), analyses the Ridge meta-learner weights, and validates the gains through paired Wilcoxon signed-rank tests. Subsection 9.3 reports the comparison against four state-of-the-art audio architectures: three pre-trained Transformers (MERT, HuBERT, wav2vec 2.0) and a Conformer trained from scratch. Subsection 9.4 closes the section with a synthesis of the findings.

9.1 Individual model performance

The evaluation of individual base models reveals substantial variability in prediction accuracy across different combinations of audio descriptors and machine learning methods. Table 2 reports the joint Mean Euclidean Distance (MED), the per-dimension Root Mean Squared Errors (RMSE_V, RMSE_A) and Mean Absolute Errors (MAE_V, MAE_A), and the per-dimension Concordance Correlation Coefficients (CCC_V, CCC_A) for all eighteen descriptor–method combinations tested in this study. The MED metric provides a holistic measure of prediction quality in the two-dimensional valence–arousal space, while CCC captures agreement between predicted and ground-truth trajectories.

Three combinations stand out at the top of the ranking and are essentially tied on the primary metric. OpenSMILE-LSTM and PAA-LSTM both achieve the lowest MED across the full table (0.241), narrowly followed by Librosa-XGBoost (0.242); the three top performers lie within 0.001 MED of each other and can be considered statistically indistinguishable on this metric at the precision considered. Their parity is confined to MED, however: on the remaining metrics OpenSMILE-LSTM separates from the other two, holding the best table-wide values for RMSE_A, MAE_A, CCC_V, and CCC_A, as discussed below. The pattern is not driven by a single dominant combination but by a small cluster of competitive ones, distributed across all three descriptor families. The bottom of the ranking is occupied by combinations involving TabNet and PLSR on PAA and Librosa, with MED

Table 2 Performance of individual descriptor–method combinations

Descriptor	Method	MED	$RMSE_V$	$RMSE_A$	MAE_V	MAE_A	CCC_V	CCC_A
OpenSMILE	LSTM	**0.241**	0.205	**0.188**	0.162	**0.146**	**0.534**	**0.754**
	MLP	0.249	0.203	0.202	0.162	0.161	0.427	0.657
	RF	0.248	0.200	0.201	0.160	0.162	0.412	0.650
	TabNet	0.263	0.218	0.214	0.173	0.168	0.421	0.655
	PLSR	0.253	0.205	0.207	0.164	0.165	0.409	0.649
	XGBoost	0.244	0.200	0.195	0.160	0.155	0.451	0.696
Librosa	LSTM	0.253	0.216	0.203	0.168	0.158	0.510	0.715
	MLP	0.248	0.204	0.201	0.163	0.160	0.430	0.662
	RF	0.243	**0.199**	0.196	**0.159**	0.156	0.440	0.681
	TabNet	0.271	0.225	0.220	0.178	0.173	0.412	0.637
	PLSR	0.250	0.205	0.202	0.164	0.161	0.400	0.669
	XGBoost	0.242	**0.199**	**0.194**	**0.159**	**0.154**	0.467	0.701
PAA	LSTM	**0.241**	0.204	0.193	0.161	0.151	0.460	0.724
	MLP	0.254	0.205	0.209	0.164	0.167	0.382	0.629
	RF	0.257	0.208	0.211	0.166	0.168	0.389	0.628
	TabNet	0.258	0.208	0.211	0.167	0.169	0.370	0.622
	PLSR	0.272	0.216	0.224	0.173	0.181	0.281	0.554
	XGBoost	0.256	0.207	0.210	0.165	0.167	0.395	0.635

Lower is better for MED, RMSE, and MAE; higher is better for CCC. Bold values mark the best result across the whole table for each metric (ties shown in bold for all tied entries)

values at or above 0.25, indicating that these paradigms are less effective at extracting predictive signal from the lower-dimensional or perceptually structured representations.

Examining performance trends across methods, no single learning paradigm dominates uniformly across the three descriptor families. LSTM achieves the lowest MED on OpenSMILE (0.241) and on PAA (0.241), while XGBoost is the best paradigm on Librosa (0.242) and the second-best on OpenSMILE (0.244). The pattern is better explained by what each paradigm exploits than by the size of the input space. LSTM is the only paradigm that consumes the frames as an ordered sequence and can model the temporal evolution of the descriptors; this gives it an advantage wherever the per-frame representation preserves a temporal structure worth tracking, which is the case both for the perceptually grounded PAA stream and for the broad OpenSMILE pool, and explains why it attains the lowest MED on these two families. On Librosa, by contrast, tree-based and boosting methods take the lead: the moderately-dimensional spectral and timbral descriptors behave as largely self-contained per-frame summaries that fit naturally with axis-aligned splits and additive boosting, so the sequential advantage of the LSTM is reduced and XGBoost and Random Forest overtake it. MLP tracks the second tier across all three families. PLSR and TabNet sit consistently in the bottom tier, with average ranks above five out of six.

When comparing descriptor families, the three produce competitive top performers and the gap between the best models across families is small. OpenSMILE hosts the lowest-MED base model (LSTM, 0.241) and the largest cluster of upper-table entries thanks to the breadth of its feature pool; Librosa contributes the strongest non-LSTM combination (XGBoost, 0.242); PAA hosts the strongest LSTM-based result on a low-dimensional representation (LSTM, 0.241). This convergence is consistent with the rationale of Sect. 5.4: the three families capture complementary informational cues, and the best method for each depends on the inductive bias of the learning paradigm rather than on a single family being uniformly superior.

Analysis of dimension-specific performance, examining MAE values separately for valence and arousal, confirms a pattern observed throughout the affective computing literature: arousal predictions exhibit lower error rates compared to valence predictions, consistent with the fact that arousal correlates more directly with acoustic energy, spectral dynamics, and rhythmic intensity, while valence depends on subtler harmonic, melodic, and timbral cues. Across the three top combinations, OpenSMILE-LSTM achieves $MAE_A = 0.146$ and $MAE_V = 0.162$; PAA-LSTM achieves $MAE_A = 0.151$ and $MAE_V = 0.161$; Librosa-XGBoost achieves $MAE_A = 0.154$

and $MAE_V = 0.159$. The CCC analysis amplifies this asymmetry: across the top combinations, CCC_A lies in the 0.70–0.75 range while CCC_V remains in the 0.46–0.53 range, reflecting the inherent difficulty of tracking valence trajectories from frame-level acoustic descriptors.

On the basis of this evidence, the consistency criterion introduced in Sect. 6.7 is applied by retaining the four learning paradigms that perform competitively across all three descriptor families: LSTM, MLP, Random Forest, and XGBoost. Paradigms are ordered by their average within-family rank, computed by ranking the six paradigms from 1 to 6 on each of the seven metrics in Table 2, separately for each descriptor family, and averaging across the resulting 21 observations per paradigm. XGBoost ranks first (1.98), narrowly ahead of LSTM (2.14), followed by MLP and Random Forest (both 3.14), with PLSR (5.21) and TabNet (5.38) trailing well behind. XGBoost ranks first by virtue of consistency rather than dominance: LSTM achieves the lowest MED on two of the three descriptor families but ranks fifth on Librosa, whereas XGBoost remains in the top three on every family. The two-position gap between the fourth and fifth paradigms provides a natural cutoff, and the twelve base models retained for the ensemble stage correspond to the four-by-three grid (LSTM, MLP, RF, XGBoost) × (OpenSMILE, Librosa, PAA), excluding TabNet and PLSR.

9.1.1 Per-feature importance

The aggregate ranking presented above identifies which descriptor–method combinations perform best, but does not address the complementary question of which specific features within each descriptor family carry the predictive content. To probe this, we report the per-feature permutation-importance analysis described in Sect. 8.6, applied to the three XGBoost base models of Table 2 (one per descriptor family). For convenience, we recall that the analysis assigns to each feature j a raw score I_j, defined as the average deterioration of MED caused by shuffling that feature on the held-out fold, and a normalised share share_j, defined as the percentage of I_j within the sum of all per-feature importances of the family. The share is the standard normalisation used in the permutation-importance literature and sums to 100% across all features of a family. The full definitions of the two quantities and the procedural details (choice of XGBoost as a unified estimator, representative-fold selection, number of permutations, sanity-check protocol) are given in Sect. 8.6; the results that follow build on that protocol without restating it.

Tables 3, 4 and 5 list the ten most important features within each family. The Importance column reports the raw score I_j defined above, while the % share column reports its normalised value within the family. The bottom row of each table reports the cumulative figures for the top-10 subset, which give a compact indicator of how concentrated the importance is within a small fraction of the pool. The same information is shown visually in Fig. 3, which extends the cumulative curve to the full feature set of each family.

The three rankings expose a consistent finding across descriptor families: the predictive content is concentrated on timbral and intensity-related features, while tonal and harmonic descriptors play a marginal role. In PAA, the

Table 3 Top-10 features by permutation importance for the PAA-Zwicker descriptor family (out of 20)

Rank	Feature	Importance	% share
1	sharp_mean	+0.042	28.47
2	loud_peak	+0.014	9.75
3	loud_mean	+0.012	8.20
4	sharp_peak	+0.011	7.80
5	rough_mean	+0.011	7.53
6	sharp_var	+0.011	7.48
7	fs_peak	+0.011	7.34
8	rough_var	+0.007	4.82
9	loud_var	+0.007	4.69
10	fs_mean	+0.005	3.75
	Top-10 total	+0.131	89.81

Importance is the raw defined above; %share is its share within the family's total positive importance. Baseline MED: 0.255. The bottom row reports the cumulative figures for the top-10 subset

Table 4 Top-10 features by permutation importance for the Librosa descriptor family (out of 77)

Rank	Feature	Importance	% share
1	mfcc_1_mean	+0.019	18.15
2	spectral_flatness_mean	+0.017	16.56
3	mfcc_1_std	+0.007	7.33
4	mfcc_2_mean	+0.007	7.00
5	spectral_rolloff_mean	+0.005	5.06
6	rms_mean	+0.004	4.36
7	spectral_flatness_std	+0.004	3.66
8	tempo	+0.004	3.57
9	rms_std	+0.003	3.25
10	spectral_contrast_3_mean	+0.002	2.26
	Top-10 total	+0.073	71.20

Importance is the raw defined above; %share is its share within the family's total positive importance. Baseline MED: 0.241. The bottom row reports the cumulative figures for the top-10 subset

Table 5 Top-10 features by permutation importance for the OpenSMILE descriptor family (out of 988)

Rank	Feature	Importance	% share
1	mfcc_sma[1]_min	+0.004	9.06
2	lspFreq_sma[0]_quartile2	+0.002	3.47
3	lspFreq_sma[0]_quartile3	+0.002	3.43
4	mfcc_sma[1]_quartile1	+0.002	3.34
5	lspFreq_sma[0]_quartile1	+0.001	2.41
6	pcm_intensity_sma_amean	+0.001	1.96
7	lspFreq_sma[7]_quartile1	+0.001	1.90
8	mfcc_sma[1]_amean	+0.001	1.90
9	lspFreq_sma[0]_amean	+0.001	1.69
10	mfcc_sma[1]_quartile3	+0.001	1.57
	Top-10 total	+0.015	30.73

Importance is the raw defined above; %share is its share within the family's total positive importance. Baseline MED: 0.245. The bottom row reports the cumulative figures for the top-10 subset

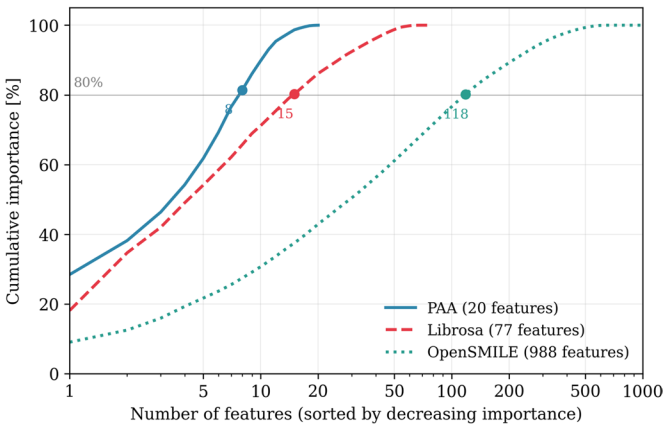


Fig. 3 Cumulative permutation-importance curves for the three descriptor families. The horizontal axis reports the number of features (logarithmic scale), ordered by decreasing importance; the vertical axis reports the cumulative share of total positive importance. Filled markers indicate where each curve crosses the 80% threshold: PAA at 8 features, Librosa at 15, OpenSMILE at 118. The horizontal separation between the three crossing points quantifies the different degrees of redundancy in each feature pool

leading feature (sharp_mean) alone accounts for 28.5% of the family's total positive importance (by far the largest single contribution observed in the analysis), and the sharpness, loudness, and roughness descriptors together account for more than 80% of the family total; tonality contributes less than 6%. In Librosa, the means of the first two MFCC coefficients and the two spectral-flatness statistics (mean and standard deviation) together account for 45% of the family total, while the twenty-four chroma features carry only about 7%. In OpenSMILE, the top-ranked features are dominated by the first MFCC coefficient and by line-spectral-pair frequencies, both of which encode the gross spectral envelope; F0 and pitch-related functionals contribute less than 1%. This convergence across three independently designed feature pools indicates that the predictive signal driving the models is largely

a timbral one, a finding that aligns with the music emotion recognition literature, where spectral brightness and noisiness are consistently identified as primary acoustic correlates of arousal.

The three families also differ markedly in how their predictive content is distributed across their feature pool, as quantified by the cumulative-importance curves of Fig. 3. PAA is the most compact: with only 20 features and a structured psychoacoustic design, the importance is spread relatively evenly, and eight features are already sufficient to capture more than 80% of the family total. Librosa is intermediate: 15 of its 77 features cover the 80% threshold, beyond which the per-feature contributions decay quickly and become negligible. OpenSMILE is the sparsest: reaching the 80% mark requires 118 of its 988 features, and the bulk of the pool contributes little individually. This pattern quantifies the cost of dimensional richness: the broader the feature pool, the larger the fraction of its features that the model treats as redundant or uninformative, which is consistent with the decision, motivated in Sect. 5, to retain the curated emobase_v2 set in full rather than applying a supervised feature selection on top of it.

Two methodological caveats apply to the interpretation of these results. First, permutation importance measures how strongly the trained model relies on each feature, not how informative the feature is in absolute terms: in the presence of correlated features the score of each is reduced because shuffling either one leaves a working surrogate available to the model. The cumulative-importance curve is therefore a conservative estimate of how much of the model's behaviour is concentrated in the top-N features, and is particularly conservative for OpenSMILE, where many functionals of the same low-level descriptor encode overlapping information. Second, the ranking is computed on the XGBoost model whose behaviour is being probed; a different learning paradigm fitted on the same feature set could distribute its importance somewhat differently.

9.2 Ensemble strategies and fusion levels

This subsection evaluates the integration of the base models at the three fusion levels introduced in Sect. 7: decision-level (late) fusion, feature-level (early) fusion, and hybrid fusion. Before turning to the fusion experiments, we make explicit the selection of the base learners that feed them. The base-model evaluation of Sect. 9.1 showed that two of the six paradigms, PLSR and TabNet, fall well below the others and, more importantly, are not competitive across all three descriptor families (their average within-family rank exceeds five out of six). Following the consistency criterion anticipated in Sect. 6.7, only the four paradigms that perform competitively on every descriptor family, namely LSTM, MLP, Random Forest, and XGBoost, are retained for the fusion stage. This yields twelve base models (four paradigms $\times$ three descriptor families), which constitute the pool from which all fusion configurations are built.

9.2.1 Late fusion

Late fusion combines the predictions of the twelve retained base models through the three decision-level strategies introduced in Sect. 7: arithmetic mean averaging, Leave-One-Fold-Out (LOFO) optimised weighted averaging, and stacking with Ridge regression meta-learning. The strategies are applied to eight configurations, instantiated by grouping the twelve base models according to complementary criteria. The configuration All 12 combines all retained base models. Configurations All LSTM, All MLP, All RF, All XGBoost combine the same learning paradigm across the three descriptor families (cross-descriptor ensembles). Configurations Within OpenSMILE, Within Librosa, Within PAA combine all four learning paradigms within a single descriptor family (cross-method ensembles).

Across configurations, a clear hierarchy emerges from Table 6 and is preserved across all three strategies: All 12 and All LSTM occupy the top of the ranking, followed by Within OpenSMILE and Within Librosa, then All XGBoost and All MLP, and finally Within PAA and All RF. This ordering is essentially the same whatever the combination mechanism, which indicates that the primary determinant of ensemble effectiveness is the diversity and quality of the constituent set rather than the sophistication of the combination strategy. The two top

Table 6 Ensemble performance across three decision-level strategies (arithmetic, LOFO, Ridge stacking) and eight configurations

Configuration	N	MED	$RMSE_V$	$RMSE_A$	MAE_V	MAE_A	CCC_V	CCC_A
All 12	12							
Arithmetic		0.228	0.189	0.181	0.151	0.144	0.485	0.724
LOFO		0.218	**0.184**	0.170	**0.146**	0.134	0.551	0.774
Ridge		**0.217**	0.184	**0.170**	0.146	**0.133**	0.549	**0.778**
All LSTM	3							
Arithmetic		0.220	0.187	0.173	0.147	0.135	**0.558**	0.776
LOFO		0.220	0.186	0.172	0.147	0.135	0.557	0.777
Ridge		0.220	0.186	0.172	0.148	0.135	0.544	0.772
W. OpenSMILE	4							
Arithmetic		0.230	0.191	0.182	0.152	0.145	0.491	0.725
LOFO		0.226	0.190	0.177	0.151	0.139	0.527	0.754
Ridge		0.225	0.189	0.176	0.150	0.139	0.518	0.760
W. Librosa	4							
Arithmetic		0.230	0.191	0.184	0.152	0.145	0.501	0.725
LOFO		0.228	0.191	0.182	0.151	0.144	0.517	0.736
Ridge		0.228	0.191	0.182	0.152	0.143	0.510	0.740
All XGBoost	3							
Arithmetic		0.237	0.195	0.190	0.156	0.151	0.456	0.700
LOFO		0.236	0.194	0.188	0.155	0.150	0.468	0.709
Ridge		0.235	0.195	0.188	0.155	0.149	0.473	0.715
All MLP	3							
Arithmetic		0.239	0.196	0.192	0.157	0.154	0.434	0.676
LOFO		0.239	0.196	0.192	0.157	0.153	0.439	0.678
Ridge		0.237	0.196	0.191	0.156	0.150	0.446	0.701
W. PAA	4							
Arithmetic		0.244	0.200	0.197	0.160	0.157	0.422	0.676
LOFO		0.238	0.199	0.189	0.158	0.149	0.453	0.717
Ridge		0.238	0.199	0.190	0.159	0.150	0.438	0.714
All RF	3							
Arithmetic		0.243	0.198	0.196	0.159	0.157	0.425	0.668
LOFO		0.241	0.197	0.194	0.158	0.155	0.437	0.679
Ridge		0.240	0.197	0.194	0.157	0.154	0.448	0.692
Best base (OS-LSTM)	1	0.241	0.205	0.188	0.162	0.146	0.534	0.754

The reference row at the bottom reports the best individual base model for comparison (OpenSMILE-LSTM). Lower is better for MED, RMSE, and MAE; higher is better for CCC. Bold marks the best result within each metric column

configurations both draw on all three descriptor families (All 12 across the four paradigms, All LSTM across the three descriptors), while the configurations confined to a single descriptor family or to a single non-LSTM paradigm trail behind, confirming that cross-descriptor diversity is the main source of ensemble gain.

The effect of the combination strategy, by contrast, is more nuanced than a simple "more sophisticated is better" progression. The decisive step is from arithmetic averaging to data-driven weighting: on the All 12 configuration the MED drops from 0.228 (arithmetic) to 0.218 (LOFO), a 4.5% reduction, as the weighting suppresses the weaker base models that an unweighted average cannot discount. The further step from LOFO to Ridge stacking, however, leaves the MED essentially unchanged (0.218 vs 0.217), and the two weighted strategies are statistically indistinguishable on this metric (Sect. 9.2.4). On the homogeneous configurations (All LSTM, Within Librosa), even the first step yields almost nothing, because the constituent models already produce highly correlated predictions and weighting has little room to improve over the simple average. The value of Ridge stacking therefore lies not in a further reduction of MED but in its dimension-specific weighting, which produces a marginal improvement in concordance (CCC_A rises from 0.774 under LOFO to 0.778 under Ridge on All 12)

and, more importantly, makes the contribution of each base model directly interpretable per emotional dimension, as analysed in Sect. 9.2.4.

Taken together, the best late-fusion result is obtained by the All 12 configuration, with LOFO and Ridge stacking essentially tied (MED = 0.218 and 0.217). Relative to the best individual base model (OpenSMILE-LSTM, MED = 0.241), the All 12 Ridge stacking reduces the MED by 9.8%, with simultaneous improvements in concordance: CCC_V rises from 0.534 to 0.549 and CCC_A from 0.754 to 0.778. This is the configuration carried forward, as the late-fusion representative, to the comparison across fusion levels and to the weight analysis.

9.2.2 Early and hybrid fusion

Late fusion combines the base models only at the level of their predictions. The two remaining fusion levels intervene earlier or at both stages. In feature-level (early) fusion, the three descriptor families are concatenated into a 1085-dimensional joint representation and one base model per retained learning paradigm is trained on it, so that cross-descriptor interactions can be exploited directly in the input space. In hybrid fusion, a Ridge stacking meta-learner combines the out-of-fold predictions of the early-fusion and late-fusion base models, integrating the two levels. Table 7 reports the seven metrics for the four early-fusion base models and for the hybrid Ridge stacking, with the best late-fusion ensemble and the best individual base model included as references.

The hybrid Ridge stacking achieves the best result of the proposed framework across all seven metrics: MED = 0.213 (a 1.8% improvement over the best late-fusion ensemble and 11.4% over the best individual base model); CCC_V = 0.575 (a 4.8% improvement over the best late-fusion ensemble and a 7.6% improvement over the best individual base model); CCC_A = 0.798 (a 2.6% improvement over the best late-fusion ensemble and a 5.8% improvement over the best individual base model). The pattern of metric improvements is consistent and reflects a real gain in tracking quality, not just in mean error magnitude.

The early-fusion base models alone are weaker than the late-fusion ensemble: their MED values lie in the 0.233–0.242 range, all worse than the late-fusion Stacking All 12 (MED = 0.217). However, the four early-fusion models follow the same paradigm ranking observed for the late-fusion base models (LSTM> XGBoost> RF > MLP), and the early-fusion LSTM trained on the 1085-dimensional concatenated representation reaches MED = 0.233, slightly better than any of the three late-fusion LSTMs trained on a single descriptor family. The early-fusion stage thus contributes complementary, not substitutive, information: it allows the LSTM to exploit cross-descriptor interactions directly in the input space, in a way that the late-fusion meta-learner can recover only indirectly through the predictions.

9.2.3 Weight analysis

The Ridge meta-learner used for late and hybrid fusion exposes, through its learned coefficients, the contribution of each base prediction to each emotional dimension. Because the meta-learner is fitted with non-negative coefficients and no intercept, the weight vector is automatically sparse and directly interpretable. We analyse first the

Table 7 Performance of the early-fusion base models and of the hybrid Ridge stacking, with reference values for the best late-fusion ensemble and the best individual base model

Experiment	MED	$RMSE_V$	$RMSE_A$	MAE_V	MAE_A	CCC_V	CCC_A
Early MLP	0.242	0.201	0.195	0.160	0.154	0.461	0.692
Early RF	0.240	0.197	0.193	0.158	0.154	0.449	0.688
Early LSTM	0.233	0.199	0.180	0.158	0.141	0.554	0.773
Early XGBoost	0.235	0.195	0.188	0.156	0.149	0.492	0.724
Hybrid ridge	**0.213**	**0.182**	**0.165**	**0.144**	**0.130**	**0.575**	**0.798**
Stacking All 12 (best late)	0.217	0.184	0.170	0.146	0.133	0.549	0.778
OpenSMILE-LSTM (best base)	0.241	0.205	0.188	0.162	0.146	0.534	0.754

Lower is better for MED, RMSE, MAE; higher for CCC. Bold marks the best within the proposed framework

Table 8 Ridge meta-learning weights for the All 12 configuration

Individual base models			Aggregated weights		
Base model	Val	Aro	Aggregate	Val	Aro
OpenSMILE-LSTM	0.282	0.328	*By learning paradigm*		
Librosa-LSTM	0.211	0.169	All LSTM (3)	0.619	0.718
PAA-LSTM	0.126	0.221	All XGBoost (3)	0.231	0.212
Librosa-XGBoost	0.169	0.126	All MLP (3)	0.122	0.057
OpenSMILE-XGBoost	0.057	0.086	All RF (3)	0.029	0.013
OpenSMILE-MLP	0.078	0.049	*By descriptor family*		
Librosa-MLP	0.041	0.008	All OpenSMILE (4)	0.419	0.463
Librosa-RF	0.026	0.013	All Librosa (4)	0.447	0.316
PAA-XGBoost	0.005	0.000	All PAA (4)	0.135	0.221
PAA-MLP	0.003	0.000			
OpenSMILE-RF	0.002	0.000			
PAA-RF	0.001	0.000			

Ridge regression (= 1.0, non-negative coefficients, no intercept) learns separate linear combinations for valence and arousal. Values are mean across folds, normalised to 1.0 within each dimension. The left side reports the weights of the twelve individual base models. The right side reports the same weights aggregated by learning paradigm (top) and by descriptor family (bottom): aggregate by paradigm sums the weights of the three base models (one per descriptor family) sharing the same learning paradigm; aggregate by descriptor sums the weights of the four base models (one per learning paradigm) sharing the same descriptor family

Table 9 Ridge meta-learning weights for the hybrid stacking. Sixteen base predictions (twelve late-fusion + four early-fusion)

Individual base predictions			Aggregated weights		
Base prediction	Val	Aro	Aggregate	Val	Aro
Early LSTM	0.242	0.322	*By learning paradigm*		
Late OpenSMILE-LSTM	0.175	0.191	All LSTM (4)	0.665	0.744
Early XGBoost	0.158	0.164	All XGBoost (4)	0.230	0.198
Late PAA-LSTM	0.104	0.156	All MLP (4)	0.096	0.058
Late Librosa-LSTM	0.144	0.075	All RF (4)	0.008	0.000
Early MLP	0.071	0.057	*By fusion level*		
Late Librosa-XGBoost	0.070	0.033	Early (4)	0.478	0.543
Late OpenSMILE-MLP	0.018	0.001	Late (12)	0.521	0.457
Early RF	0.007	0.000			
Late Librosa-MLP	0.004	0.000			
Late PAA-MLP	0.003	0.000			
Late OpenSMILE-XGBoost	0.002	0.001			
Late Librosa-RF	0.001	0.000			
Late OpenSMILE-RF	0.000	0.000			
Late PAA-RF	0.000	0.000			
Late PAA-XGBoost	0.000	0.000			

Values are mean across folds, normalised to 1.0 within each dimension. The left side reports the weights of the sixteen individual base predictions. The right side reports the same weights aggregated by learning paradigm (top, summing early- and late-fusion predictions sharing the same paradigm) and by fusion level (bottom, summing all four early-fusion and all twelve late-fusion predictions)

late-fusion All 12 configuration (Table 8), then the hybrid configuration (Table 9); the two are visualised in panels (a) and (b) of Fig. 4.

The weight table exposes a clear and interpretable structure. LSTM-based models receive the largest aggregate share of the Ridge weights, accounting for 0.619 of the valence prediction and 0.718 of the arousal prediction in the All 12 configuration. The contribution is distributed across all three descriptor families: OpenSMILE-LSTM is the single largest contributor (0.282 valence, 0.328 arousal), followed by PAA-LSTM (0.126 valence, 0.221 arousal) for arousal and by Librosa-LSTM (0.211 valence, 0.169 arousal) for valence. The temporal modelling capacity of LSTM thus dominates the ensemble irrespective of the input descriptor, a pattern that aligns with the predictive importance of temporal dependencies in continuous emotion tracking.

XGBoost emerges as the second contributing paradigm, with an aggregate share of 0.231 for valence and 0.212 for arousal. The weight allocation is concentrated on the Librosa and OpenSMILE families (Librosa-XGBoost:

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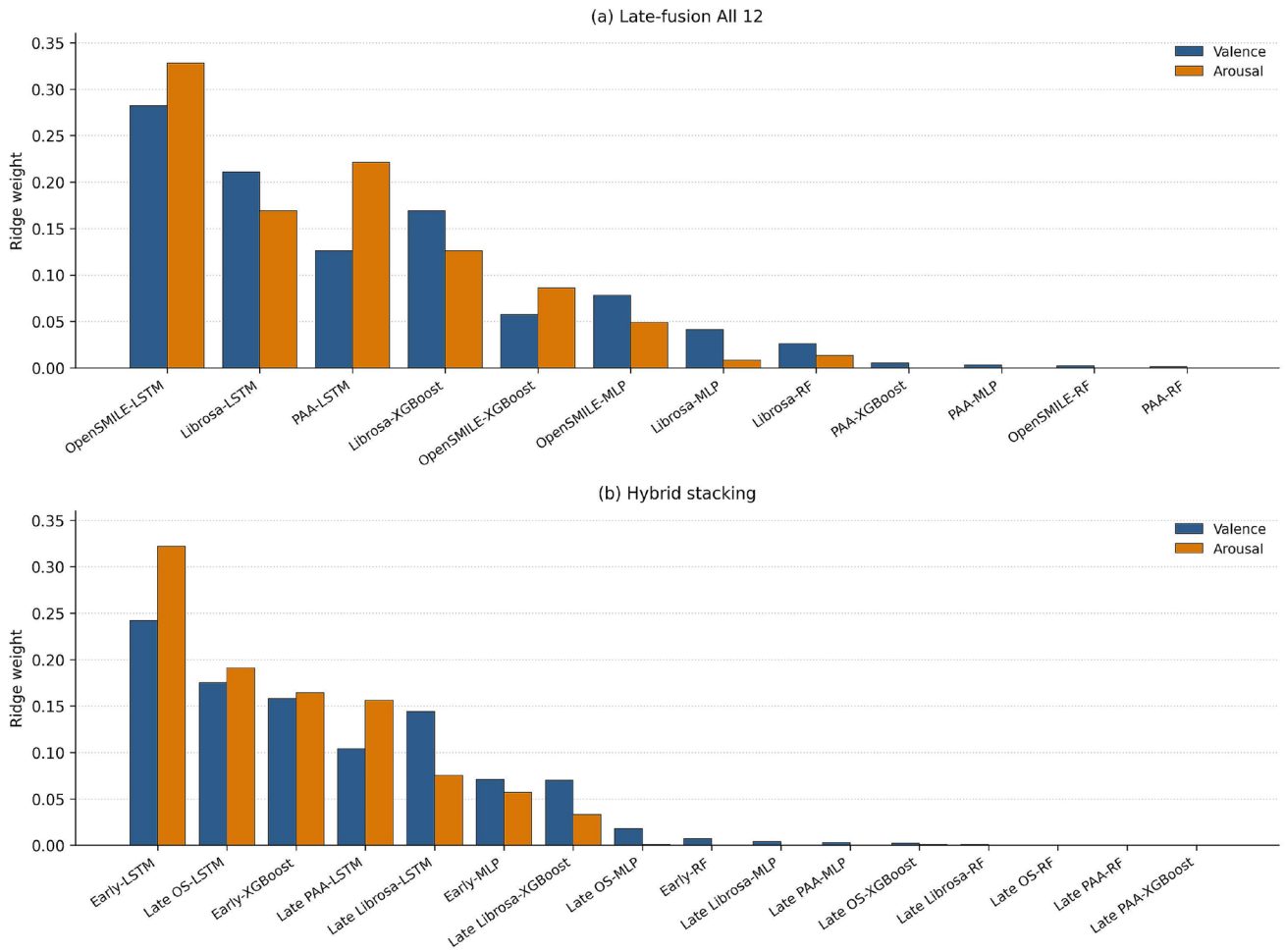


Fig. 4 Ridge meta-learner weights for (**a**) the late-fusion All 12 configuration (Table 8) and (**b**) the hybrid stacking configuration (Table 9). For each base prediction the figure reports two bars, valence (blue) and arousal (orange); within each panel, base predictions are ordered by the mean of their valence and arousal weights in descending order. Weights are mean across folds and normalised to 1.0 within each dimension

0.169 valence, 0.126 arousal; OpenSMILE-XGBoost: 0.057 valence, 0.086 arousal), while the contribution from PAA-XGBoost is essentially zero. The pattern reflects the inductive bias of gradient boosting: tree-based models thrive on the higher-dimensional Librosa and OpenSMILE inputs but offer little additional signal on the perceptually structured PAA representation, where the LSTM already extracts most of the available information.

MLP and Random Forest contribute marginally (aggregate shares of 0.122 and 0.029 for valence, 0.057 and 0.013 for arousal). Half of their six base models receive weights at or below 0.01 on both dimensions: the non-negative Ridge formulation performs an automatic selection, retaining only the predictions that add content beyond what LSTM and XGBoost already provide and discarding the redundant ones.

Examining the dimension-specific patterns, the descriptor-aggregated weights in the right side of Table 8 expose a clean separation. The Librosa family contributes substantially more to valence than to arousal (0.447 vs 0.316), the largest dimension-asymmetry observed in the table. This is consistent with the harmonic and chromatic content captured by Librosa descriptors (chroma features, spectral centroid, MFCCs), which carry information relevant to harmonic mode and timbral character, the acoustic correlates traditionally associated with valence. Symmetrically, the PAA family contributes more to arousal than to valence (0.221 vs 0.135), and the OpenSMILE family also leans toward arousal (0.463 vs 0.419). This reflects the energy-, loudness-, and roughness-based content that dominates these descriptor families, and that aligns with the acoustic correlates of

arousal documented in the psychoacoustic literature. The dimension-specific weighting of the Ridge meta-learner thus exposes a clean separation that is theoretically expected and empirically robust across folds.

This complementarity is the structural reason why the hybrid stacking outperforms both fusion levels in isolation. Table 9 reports the Ridge weights learned by the hybrid meta-learner over the pool of sixteen base predictions (twelve late-fusion + four early-fusion), separately for valence and arousal; the corresponding visualisation is panel (b) of Fig. 4.

Three patterns dominate the hybrid weight structure. First, the early-fusion and late-fusion contributions are nearly balanced, with early-fusion accounting for 0.478 of the valence prediction and 0.543 of the arousal prediction, and late-fusion accounting for the remainder. The meta-learner thus does not select one fusion level over the other but rather combines them in roughly equal proportions, confirming that the two carry complementary information. Second, LSTM dominates the contribution as in the late-fusion All 12 ensemble: aggregating the four LSTM models (one early plus three late) gives a share of 0.665 for valence and 0.744 for arousal. This is consistent with the corresponding aggregate in the late-fusion Ridge weights (0.619/0.718 in Table 8) and with the literature consensus that temporal modelling is essential for continuous emotion tracking. Third, eight out of sixteen base predictions receive a weight at or below 0.01 in both dimensions and are effectively discarded by the meta-learner: the four Random Forest variants (Early RF and the three Late RFs), the two PAA-MLP and Librosa-MLP late predictions, and the two PAA-XGBoost and OpenSMILE-XGBoost late predictions. The non-negative Ridge constraint thus performs an automatic selection on the input pool, retaining only the predictions that add informational content.

9.2.4 Statistical significance

The gains achieved by the fusion strategies are assessed with paired Wilcoxon signed-rank tests on the per-fold MED values, following the recommendation of Demšar [53] for the non-parametric comparison of learning algorithms across matched folds. With $n = 5$ folds, the strongest attainable one-sided signal is $W = 15$ at $p = 0.031$, obtained when one system has a lower per-fold error than the other on all five folds. The decisive step in the late-fusion hierarchy passes this threshold: both the LOFO and the Ridge stacking All 12 ensembles achieve a lower MED than the arithmetic All 12 ensemble on every one of the five folds ($W = 15$, $p = 0.031$). The further step from LOFO to Ridge stacking, by contrast, does not reach significance ($W = 11$ on five folds, $p \approx 0.31$): the two weighted strategies are statistically indistinguishable on MED, consistent with their near-identical mean values. The hybrid Ridge stacking, in turn, improves on the best late-fusion ensemble on all five folds, recovering the maximal $W = 15$, $p = 0.031$ signal. The same maximal signal separates the late-fusion ensemble from each of the four early-fusion base models, which it outperforms on every fold ($W = 15$, $p = 0.031$ in all four comparisons), confirming that the hybrid gain comes from genuine complementarity rather than from the early-fusion models being competitive on their own. The picture that emerges is therefore one of two statistically validated steps, from unweighted to weighted late fusion and from late to hybrid fusion, separated by a plateau in which the choice between LOFO and Ridge stacking is, for predictive accuracy, immaterial; the reason to prefer Ridge stacking is its dimension-specific interpretability rather than a significant accuracy gain.

9.3 Comparison with state-of-the-art audio architectures

To position the proposed framework against the current state of the art in audio representation learning, four state-of-the-art audio architectures are evaluated on the same five cross-validation folds. Three are pre-trained Transformers used under the unified probing protocol described in Sect. 8: MERT [23], pre-trained on music; HuBERT [22], pre-trained on speech with a masked discrete-target prediction objective; and wav2vec 2.0 [21], pre-trained on speech with a contrastive masked-prediction objective. The fourth is a Conformer [20] trained from scratch on the DEAM mel-spectrograms, included as an architectural baseline that does not rely on external data. Table 10

Table 10 Performance of the four state-of-the-art audio architectures and of the proposed framework

Model	MED	$RMSE_V$	$RMSE_A$	MAE_V	MAE_A	CCC_V	CCC_A
MERT (music)	**0.201**	**0.167**	**0.162**	**0.132**	**0.127**	**0.680**	**0.810**
HuBERT (speech)	0.236	0.192	0.190	0.153	0.151	0.551	0.733
wav2vec 2.0 (speech)	0.239	0.197	0.192	0.156	0.152	0.519	0.718
Conformer (from scratch)	0.246	0.202	0.195	0.162	0.157	0.457	0.705
Hybrid Ridge (ours, best)	0.213	0.182	0.165	0.144	0.130	0.575	0.798
Stacking All 12 (ours, best late)	0.217	0.184	0.170	0.146	0.133	0.549	0.778
OpenSMILE-LSTM (ours, best base)	0.241	0.205	0.188	0.162	0.146	0.534	0.754

The first three are pre-trained Transformers; the fourth is a Conformer trained from scratch. The lower block reports the proposed framework, with the hybrid Ridge stacking as its best configuration. Lower is better for MED, RMSE, and MAE; higher is better for CCC. Bold marks the best result per metric

reports the resulting performance, alongside the three reference configurations of the proposed framework: the hybrid Ridge stacking (its best configuration), the best late-fusion ensemble, and the best individual base model.

Four observations emerge. First, MERT achieves the lowest error of any model evaluated (MED = 0.201), with CCC values of 0.680 for valence and 0.810 for arousal that exceed those of every other model in this study. The result reflects the advantage of music-domain self-supervised pre-training: a 330M-parameter encoder trained on a large unlabelled music corpus produces representations intrinsically adapted to the acoustic structure of the task. The best configuration of the proposed framework, the hybrid Ridge stacking (MED = 0.213), is the closest competitor to MERT, trailing it by 6.0% in MED, down from the 8.0% gap of the best late-fusion ensemble. The hybrid framework thus narrows the distance to the music-pre-trained Transformer while relying on no external data and a fraction of the parameters. Second, the speech-domain Transformers (HuBERT and wav2vec 2.0) trail the proposed framework: HuBERT achieves MED = 0.236 and wav2vec 2.0 MED = 0.239, both above the hybrid Ridge stacking (0.213) and the best late-fusion ensemble (0.217). The transfer of speech-pre-trained encoders to music emotion recognition is therefore beneficial relative to most individual base models but not sufficient to match the proposed ensemble of hand-crafted descriptors. Third, the gap between MERT and HuBERT (the same family of architectures pre-trained on different corpora) is substantially larger than the gap between HuBERT and wav2vec 2.0 (different architectures pre-trained on similar corpora), indicating that for this task the pre-training domain matters more than the specific encoder design. Fourth, the Conformer trained from scratch reaches MED = 0.246, below all other state-of-the-art baselines and the only one that also trails the best individual base model. The proposed hybrid framework outperforms it by 13.4% in MED (0.213 vs 0.246), a directly apples-to-apples comparison since neither model relies on external data. The architecture alone, without large-scale pre-training, is not sufficient on a corpus of the size of DEAM, and the gap to the pre-trained Transformers quantifies the contribution of pre-training in isolation from architectural design.

Beyond raw accuracy, two considerations qualify the comparison. First, MERT operates with two orders of magnitude more parameters than the proposed framework and requires substantially more compute at inference time (feature extraction on the order of hours of CPU time for the dataset, versus minutes for the hand-crafted descriptors). The hybrid framework remains the practical choice for applications with latency or compute constraints, while closing most of the accuracy gap. Second, the meta-learner weights of the proposed framework provide direct interpretability of the contributions of each descriptor family, learning paradigm, and fusion level to each emotional dimension, an analysis that is unavailable for the Transformer baselines.

9.4 Remarks

Three broader observations emerge from the experiments presented in this section.

The first concerns where the predictive signal actually resides. Across base models, ensembles, and fusion levels, the gains track the diversity of the information being combined far more closely than the sophistication of the mechanism combining it. Weighting beats averaging, hybrid beats single-level fusion, and the All 12 and

All LSTM configurations lead the ranking, but in every case the improvement materialises only when genuinely different acoustic views are brought together, and saturates as soon as the combination becomes homogeneous. The corollary is practical: for continuous MER, effort is better spent broadening the descriptor and fusion repertoire than refining a single combiner, since the meta-learner stage contributes a real but bounded increment once diversity is in place.

The second concerns interpretability as a structural property rather than an add-on. Because every combination stage in the framework is a linear meta-learner, the same coefficients that produce the prediction also explain it, and they do so in a way that recovers a theoretically expected structure without being told to: Librosa weight concentrates on valence, OpenSMILE and PAA on arousal, and temporal models dominate both. This alignment between learned weights and psychoacoustic expectation is the kind of evidence that a black-box encoder cannot supply, and it is what distinguishes the contribution of this framework from a mere accuracy number.

The third concerns the position of hand-crafted-descriptor ensembles in the current landscape. The comparison with pre-trained Transformers isolates pre-training domain, not architecture, as the decisive variable: a transparent ensemble of classical descriptors surpasses speech-pre-trained Transformers and a from-scratch Conformer, and trails only the music-pre-trained MERT, which it approaches to within a narrow margin without any external data. The practical implication is that, where compute, data, or interpretability are constraints, an interpretable descriptor ensemble remains a competitive and fully transparent alternative, and a natural baseline against which future music-pre-trained or hybrid systems can be measured.

10 Conclusions

This study presented a comprehensive comparative framework for predicting continuous emotional trajectories in music using the valence–arousal model of affect. Through systematic evaluation of three theoretically diverse audio descriptor families (OpenSMILE, PAA, Librosa) combined with six complementary machine learning paradigms (LSTM, MLP, Random Forest, TabNet, PLSR, XGBoost), we investigated how descriptor diversity, model architecture, and integration strategy jointly contribute to emotion prediction accuracy. The integration was studied at the three canonical fusion levels: decision-level (late) through arithmetic averaging, LOFO-weighted averaging, and Ridge stacking; feature-level (early) through descriptor concatenation; and hybrid stacking that combines early- and late-fusion predictions. The proposed framework was further benchmarked against four state-of-the-art audio architectures: three pre-trained Transformers (MERT, HuBERT, wav2vec 2.0) under a unified probing protocol, and a Conformer trained from scratch on the DEAM mel-spectrograms. The ensemble gains were validated through paired Wilcoxon signed-rank tests, and the predictive content of each descriptor family was further characterised through a per-feature permutation-importance analysis. The experimental design prioritised methodological rigour through file-wise cross-validation to prevent data leakage, unified preprocessing pipelines, and comprehensive evaluation using a primary multi-output metric (Mean Euclidean Distance, MED) supported by dimension-specific RMSE, MAE, and Concordance Correlation Coefficient (CCC). Beyond quantitative performance metrics, this work provides empirically validated insights into the relative strengths of different feature representations and modelling approaches, establishing a transparent and reproducible baseline for comparative MER research.

The six research questions posed in Sect. 4 receive direct empirical answers through the experimental results of Sect. 9.

First, regarding descriptor families (Table 2 of Sect. 9): the three families OpenSMILE, Librosa, and PAA produce competitive top performers and the gap between the best models across families is small. OpenSMILE-LSTM and PAA-LSTM both achieve the lowest MED of any individual base model (0.241), narrowly ahead of Librosa-XGBoost (0.242); the three top performers lie within 0.001 MED of each other. Each descriptor family hosts one of these top three combinations, indicating that the families capture complementary informational cues: OpenSMILE excels when paired with sequence learning on its broad affective-computing pool, PAA hosts

the strongest LSTM-based result on a low-dimensional perceptually structured representation, and Librosa contributes the strongest non-LSTM combination through gradient boosting on its moderately-dimensional MIR descriptors. The choice of the most effective descriptor depends on the inductive bias of the learning paradigm with which it is paired, rather than on a single family being uniformly superior. The complementary per-feature permutation-importance analysis reported in Sect. 9.1.1 sharpens this picture: across all three families the predictive content is concentrated on timbral and intensity-related descriptors (sharpness and loudness in PAA, the first MFCC coefficients and spectral flatness in Librosa, the first MFCC coefficient and line-spectral-pair frequencies in OpenSMILE), while tonal and harmonic descriptors play a marginal role in each family. The three families differ, however, in how concentrated their predictive content is: the top-10 features cover 89.8% of the total positive importance in PAA, 71.2% in Librosa, and only 30.7% in OpenSMILE, quantifying the redundancy that characterises broader feature pools.

Second, regarding modelling paradigms (Table 2 of Sect. 9): two paradigms, LSTM and XGBoost, stand out as the strongest, but neither is best on every descriptor family. LSTM achieves the lowest MED on OpenSMILE (0.241) and on PAA (0.241), while XGBoost is the strongest paradigm on Librosa (0.242) and the second-best on OpenSMILE (0.244). The discriminating factor is what each paradigm exploits rather than the size of the input space: LSTM is the only paradigm that consumes the frames as an ordered sequence, and it leads wherever the per-frame representation preserves a temporal structure worth tracking, as in the PAA and OpenSMILE streams; on Librosa, whose descriptors behave as largely self-contained per-frame summaries, axis-aligned splits and additive boosting take the lead instead. The average within-family rank, computed across the seven metrics of Table 2 (21 observations per paradigm), identifies a clear cutoff after the fourth paradigm: XGBoost (1.98), LSTM (2.14), MLP and Random Forest (both 3.14) form a top tier, while PLSR (5.21) and TabNet (5.38) trail by over two rank positions. The four top paradigms are thus retained as the base learners for the ensemble stage, while PLSR and TabNet are excluded by the consistency criterion introduced in Sect. 6.7.

Third, regarding decision-level ensemble strategies (Table 6 of Sect. 9): combining the twelve retained base models through Ridge meta-learning achieves the best late-fusion performance, with the All 12 configuration reaching MED = 0.217, $CCC_V = 0.549$, and $CCC_A = 0.777$. This corresponds to a 9.8% MED improvement over the best individual base model (OpenSMILE-LSTM, MED = 0.241), a 2.7% improvement in CCC_V (0.534 to 0.549), and a 3.2% improvement in CCC_A (0.754 to 0.777). Progressive improvements in MED are observed from arithmetic averaging (0.228) to LOFO-weighted averaging (0.218) to Ridge stacking (0.217), with the largest single jump occurring in the move from arithmetic to LOFO weighting; the further refinement from LOFO to Ridge stacking is marginal in MED but produces measurable gains in CCC. The paired Wilcoxon signed-rank tests reported in Sect. 9.2 validate the two substantive steps of the hierarchy, from unweighted to weighted late fusion and from late to hybrid fusion, at the maximum attainable significance level ($W = 15$, $p = 0.031$ on five folds), while the refinement from LOFO to Ridge stacking is not statistically significant on MED.

Fourth, regarding fusion levels (Table 7 of Sect. 9): the hybrid Ridge stacking, combining the predictions of the late-fusion base models and the early-fusion base models, achieves the best result of the proposed framework (MED = 0.213, $CCC_V = 0.575$, $CCC_A = 0.798$). The hybrid result improves on the best late-fusion ensemble by 1.8% in MED, 4.8% in CCC_V, and 2.6% in CCC_A, and improves on the best individual base model by 11.4% in MED, 7.6% in CCC_V, and 5.8% in CCC_A. The early-fusion base models alone are weaker than the late-fusion ensemble (MED in the 0.233–0.242 range), but the hybrid meta-learner extracts complementary signal from both fusion levels, with weights distributed in roughly equal proportions (47.8% early, 52.2% late for valence; 54.3% early, 45.7% late for arousal; Table 9 of Sect. 9). The result confirms that the locus of integration matters and that decision-level and feature-level fusion carry partially non-overlapping information.

Fifth, regarding interpretability (Tables 8 and 9, and the per-feature analysis of Sect. 9.1.1): the framework exposes its predictive content at two complementary levels of granularity. At the family and paradigm level, the Ridge meta-learner weights show that LSTM-based models receive the largest aggregate share (61.9% of the valence prediction and 71.8% of the arousal prediction in the late-fusion All 12 configuration, 66.5%/74.4% in the hybrid stacking), with XGBoost second at roughly 23% in both fusion levels. The weights also expose a clean

dimension-specific separation: the Librosa family contributes more to valence than to arousal, while OpenSMILE and PAA contribute proportionally more to arousal, an asymmetry that aligns with the psychoacoustic correlates of the two emotional dimensions. At the individual-feature level, the permutation-importance analysis concentrates the predictive content on timbral and intensity-related descriptors across all three families, with markedly different concentration (the top-10 features cover 89.8% of the total positive importance in PAA, 71.2% in Librosa, and 30.7% in OpenSMILE). This two-level interpretability, unavailable for the Transformer baselines, is the distinctive methodological contribution of the framework.

Sixth, regarding the comparison against state-of-the-art audio architectures (Table 10 of Sect. 9): under a unified probing protocol with frozen encoder, multi-layer weighted-sum aggregation, and an MLP probe head, the music-domain pre-trained MERT achieves the lowest error of any model evaluated (MED = 0.201, CCC_V = 0.680, CCC_A = 0.810). The proposed hybrid Ridge stacking remains its closest competitor, within 6.0% in MED while relying on no external data and a fraction of the parameters. The two speech-domain pre-trained Transformers trail the proposed framework (HuBERT at MED = 0.236 and wav2vec 2.0 at MED = 0.239, both above the hybrid Ridge stacking at 0.213 and the late-fusion ensemble at 0.217), and the Conformer trained from scratch trails further (MED = 0.246), indicating that transfer from speech and architecture alone are beneficial relative to most individual base models but not sufficient to match an interpretable hand-crafted-descriptor ensemble. The pre-training domain is the dominant factor explaining the gap, with the encoder architecture playing a secondary role: the gap between MERT and HuBERT (same family of architectures, different domains) is substantially larger than the gap between HuBERT and wav2vec 2.0 (different architectures, similar domains).

Beyond the empirical findings, this work establishes a methodological reference for affective computing research. The framework's modularity (separate stages for feature extraction, base-model training, and ensemble combination at three fusion levels), its reliance on publicly available libraries and standardised feature extraction tools, and its validation through paired Wilcoxon signed-rank tests jointly ensure full reproducibility and provide a consistent baseline for future investigations. A distinctive feature is the interpretability of the meta-learner weights, which exposes how individual descriptors and paradigms contribute to each emotional dimension and supports applications where transparent predictions are required, such as music therapy or recommendation systems.

Several limitations warrant acknowledgement. The DEAM dataset, while widely adopted, comprises primarily Western music and 45-second excerpts, which may limit generalisation to complete musical works, non-Western styles, and longer-term emotional arcs. The relatively modest inter-rater reliability (Krippendorff's alpha around 0.20) reflects the inherent subjectivity of emotion perception, meaning that aggregated annotations represent collective tendencies rather than ground truth. Finally, the deliberate focus on established methods rather than state-of-the-art deep architectures prioritises interpretability and reproducibility over absolute performance, leaving room for further accuracy gains through more complex models or domain-adaptive transfer from large pre-trained audio encoders.

Future research will extend this framework in three main directions. Cross-dataset generalisation is a natural first step: evaluating the framework on a second continuous valence–arousal corpus such as PMEmo would test whether the observed contribution patterns and fusion-level hierarchy transfer to different annotation protocols and musical populations. Domain-adaptive transfer from music-pre-trained encoders such as MERT, possibly combined with the proposed hybrid framework, may close the residual gap to MERT while preserving interpretability. Finally, multimodal integration of lyrics, metadata, and physiological signals (EDA, EEG, heart rate) would enable a more complete account of musical affective experience.

In conclusion, this work demonstrates that systematic comparison, methodological rigour, and transparent evaluation are as valuable as architectural innovation in advancing Music Emotion Recognition research, and provides an empirically validated baseline that balances accuracy, interpretability, and reproducibility for future studies.

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Data availability The work uses already publicly available datasets and materials. References are appropriately cited in the paper.

Declarations

Competing interests No competing interests for this work.

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